

“NOXIOUS NOTES”

HOW TO BE A DBA

Database Administrator.



To everyone accessing this document,
This is the result of my journey and notes as a database administrator. Whether you are an engineer or an aspiring one, I hope you enjoy and make good use of these notes. Spread kindness by helping others, by supporting each other to grow.

There's a quote from Bob Sadino that says:

"When you meet someone new, don't forget to empty your glass."

This is the principle I've held onto so far.

Happy reading and growing, fellow engineers!

— noxiousroots a.k.a Bimo Anggoro Jati, S.Kom

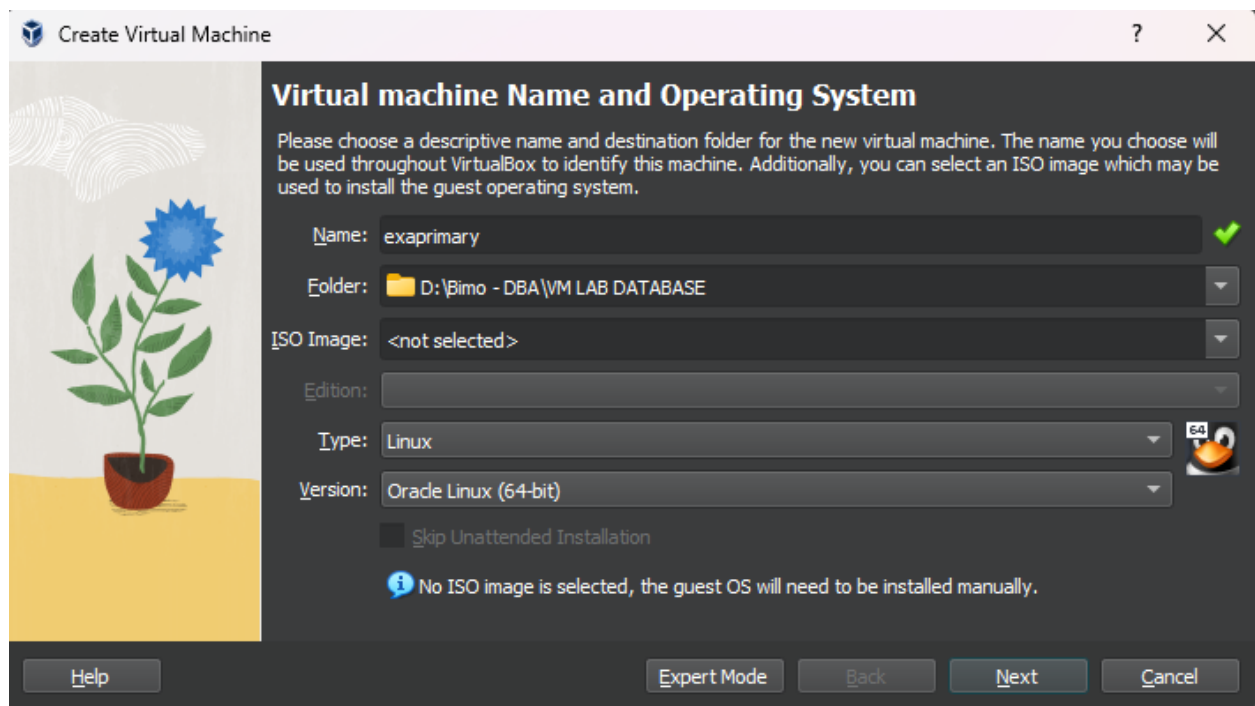
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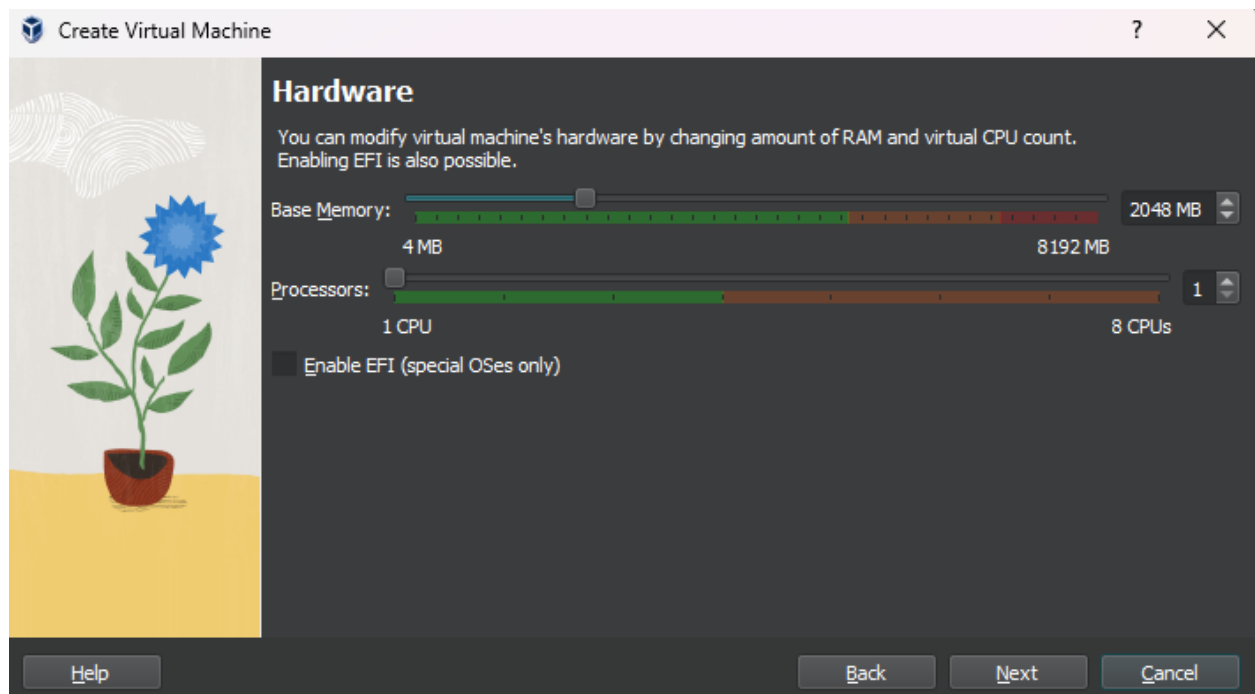
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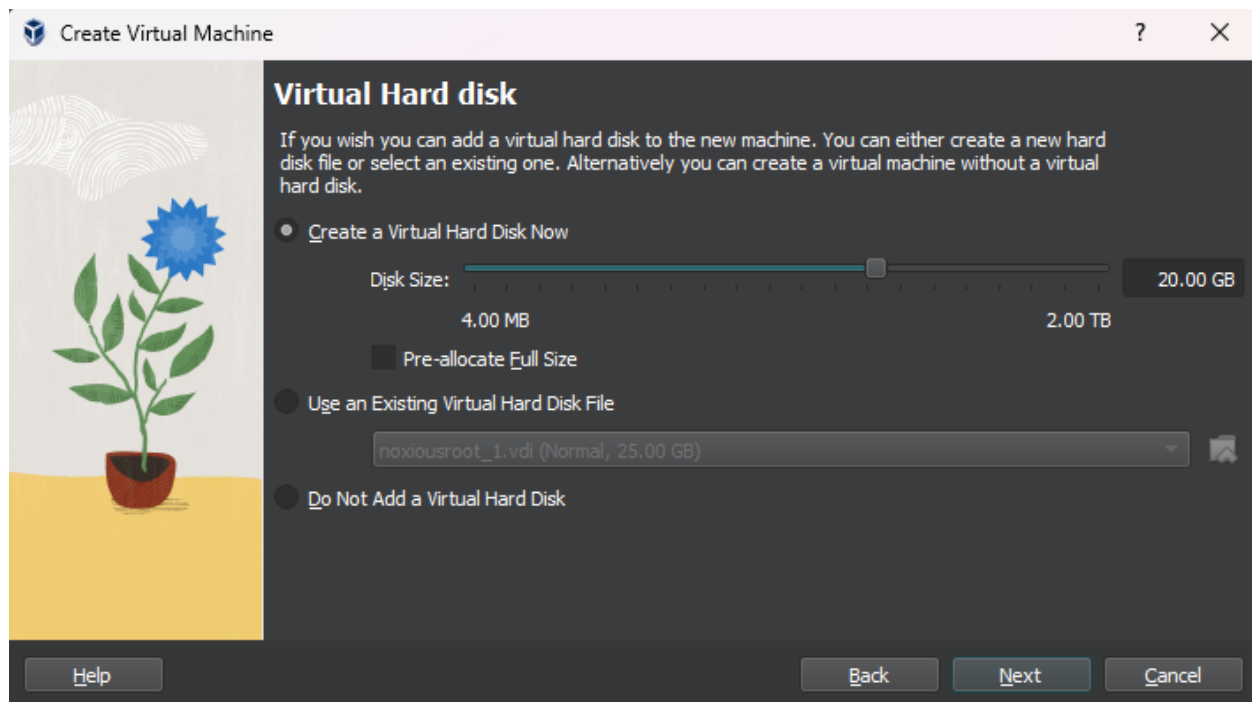
INSTAL ORACLE LINUX 7 ON VIRTUAL MACHINE



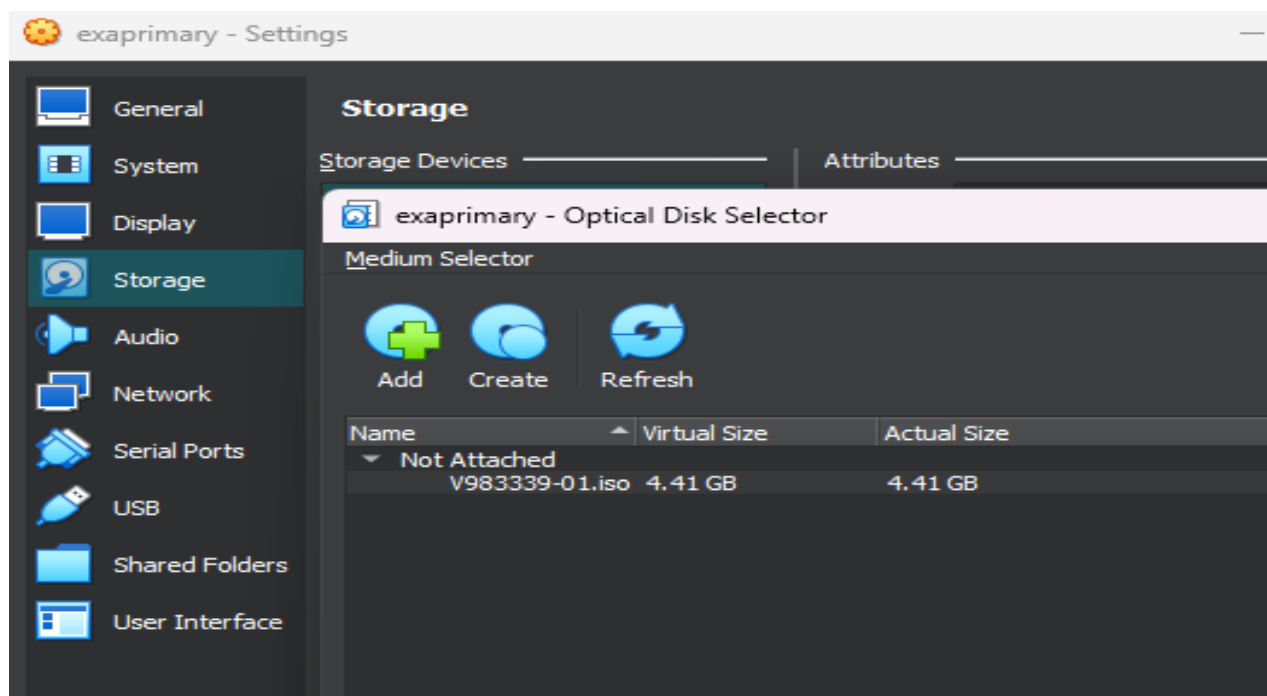
Set nama untuk server yang akan di install, dan menentukan path Dimana folder akan disimpan.



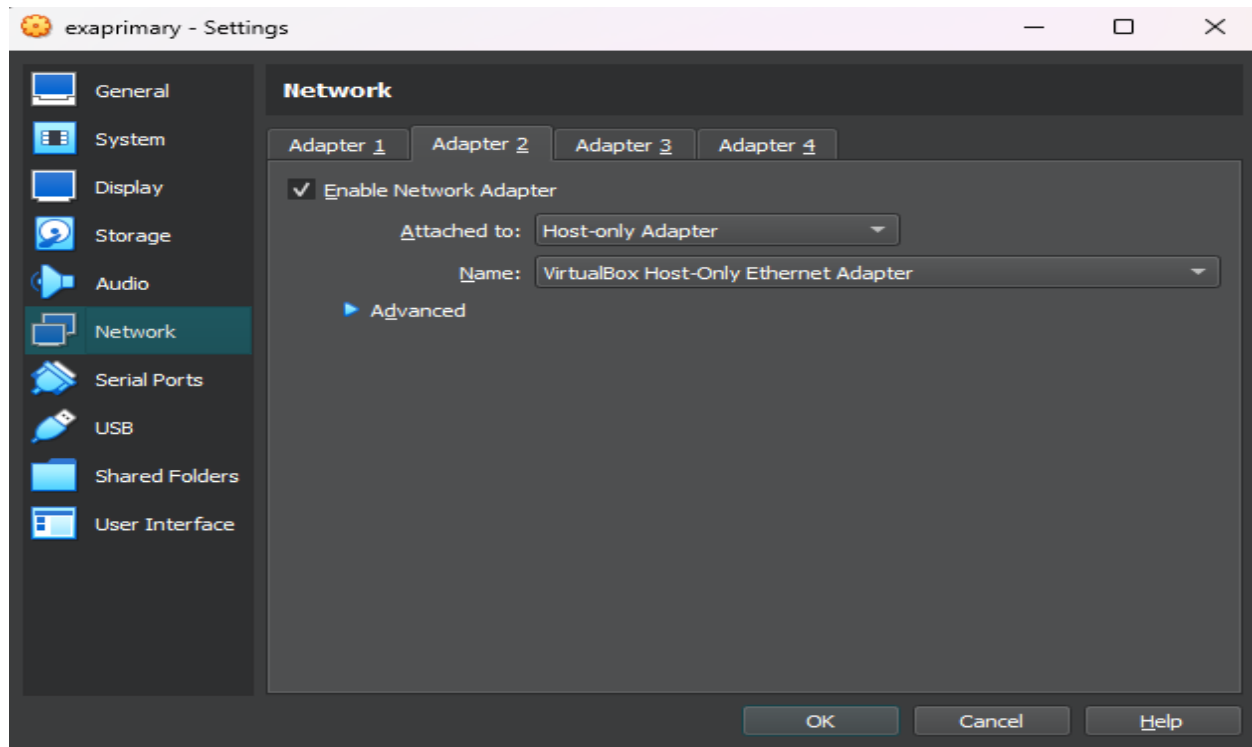
Set base memory (RAM) dan processor yang akan digunakan untuk server. (OPTIONAL)



Set virtual hard disk yang akan di alokasikan sebagai penyimpanan server. (OPTIONAL)



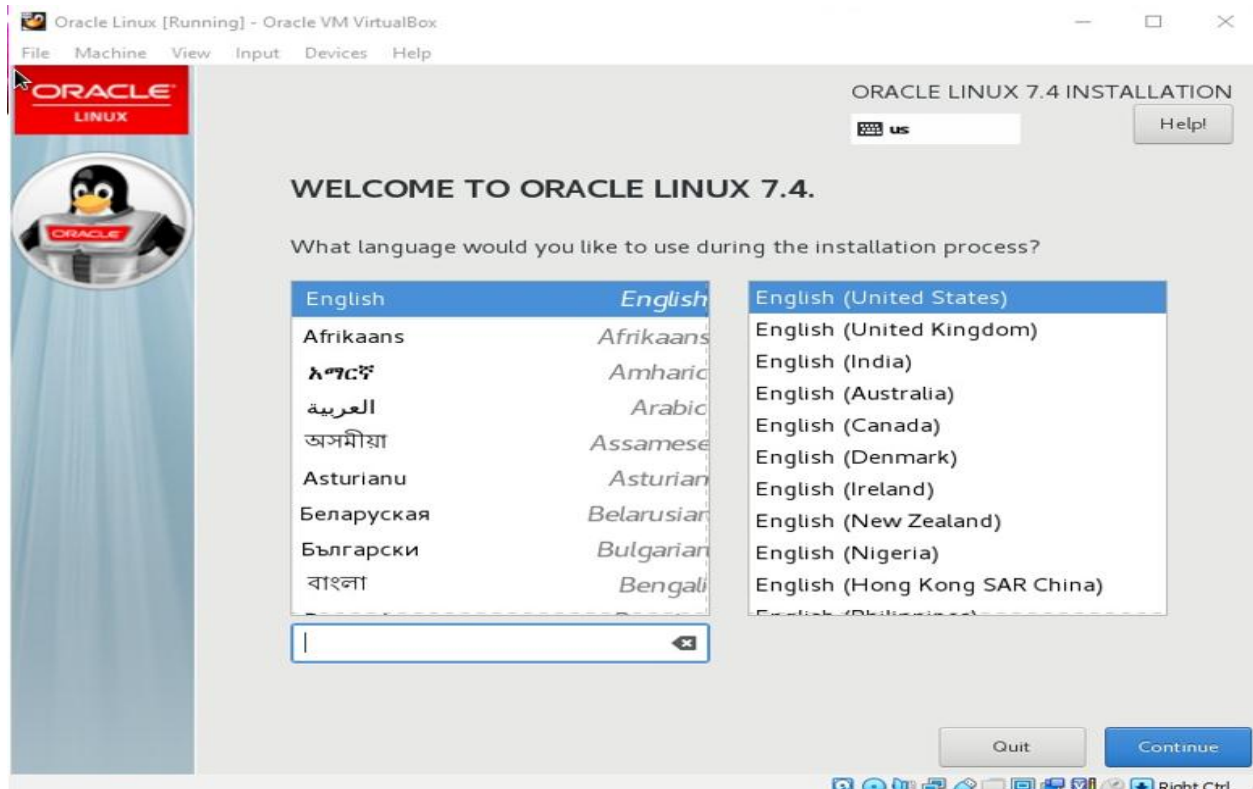
Add iso oracle linux 7 yang sudah di download pada settingan server exaprimary.



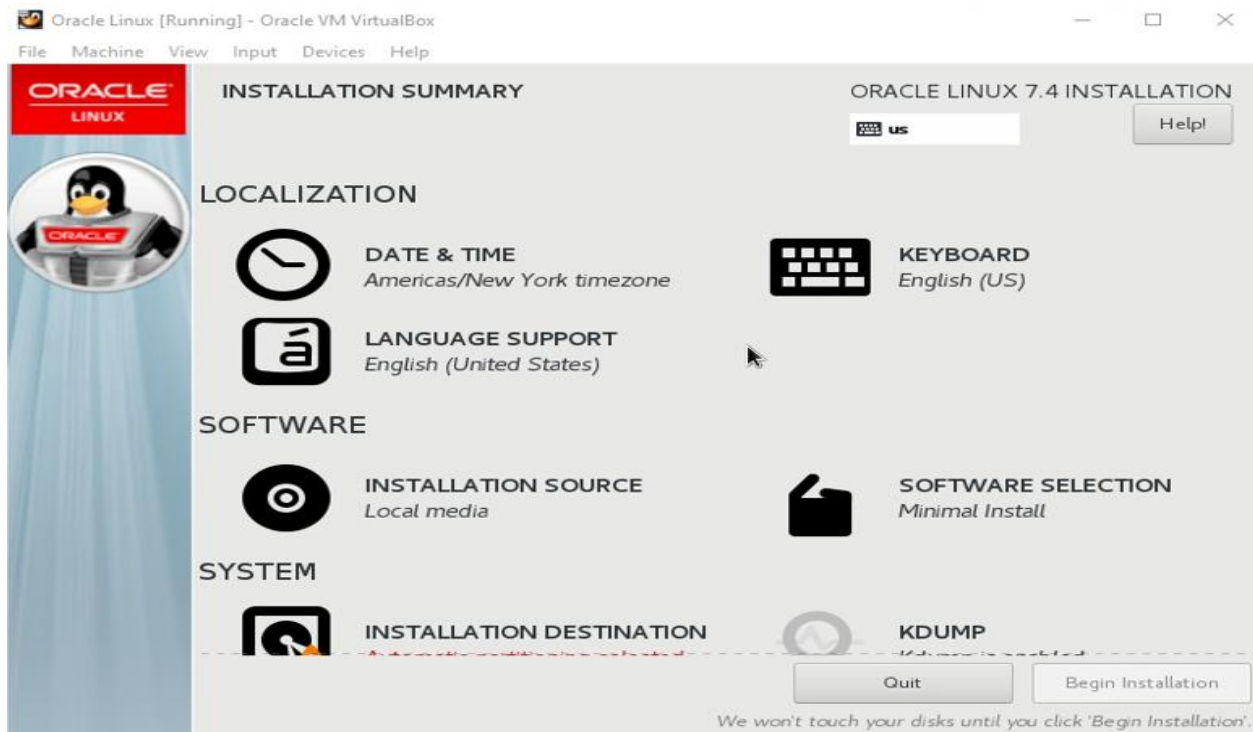
Enable adapter 2 dan set ke Host-only Adapter, untuk keperluan SSH dari MobaXterm.



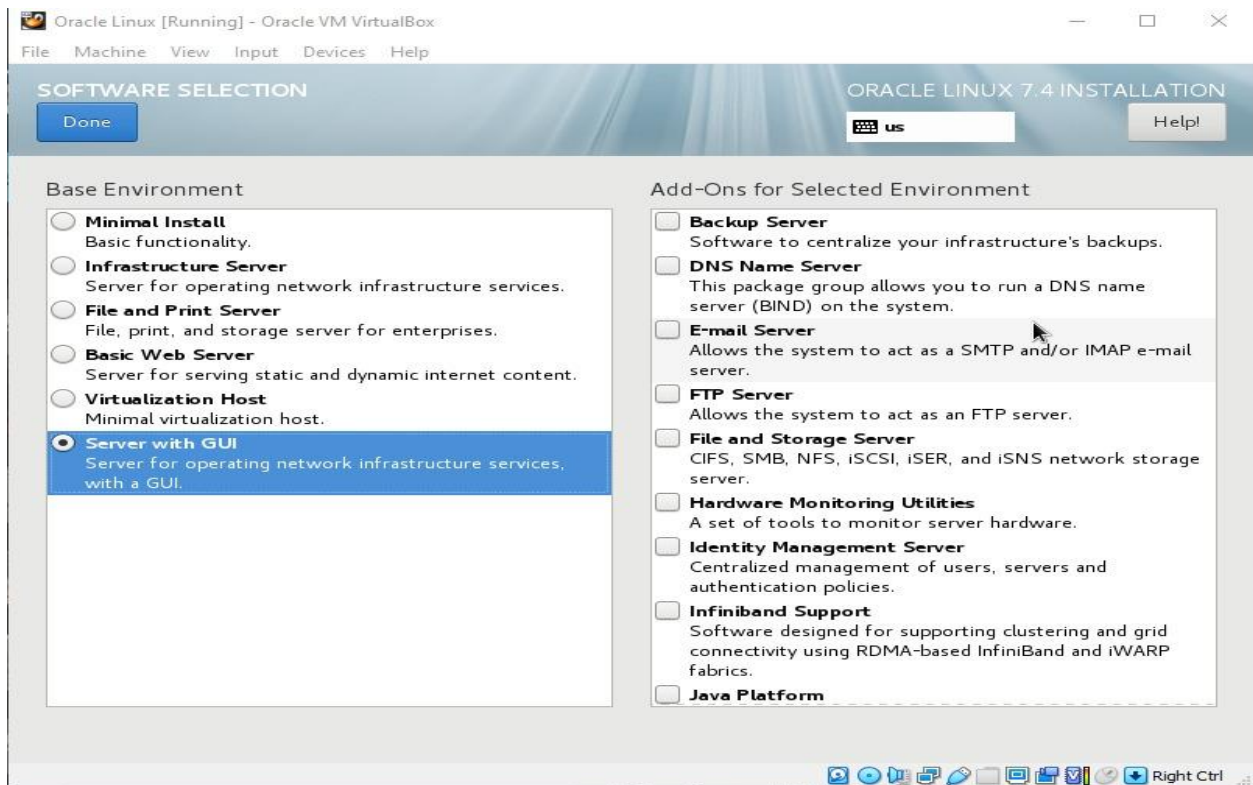
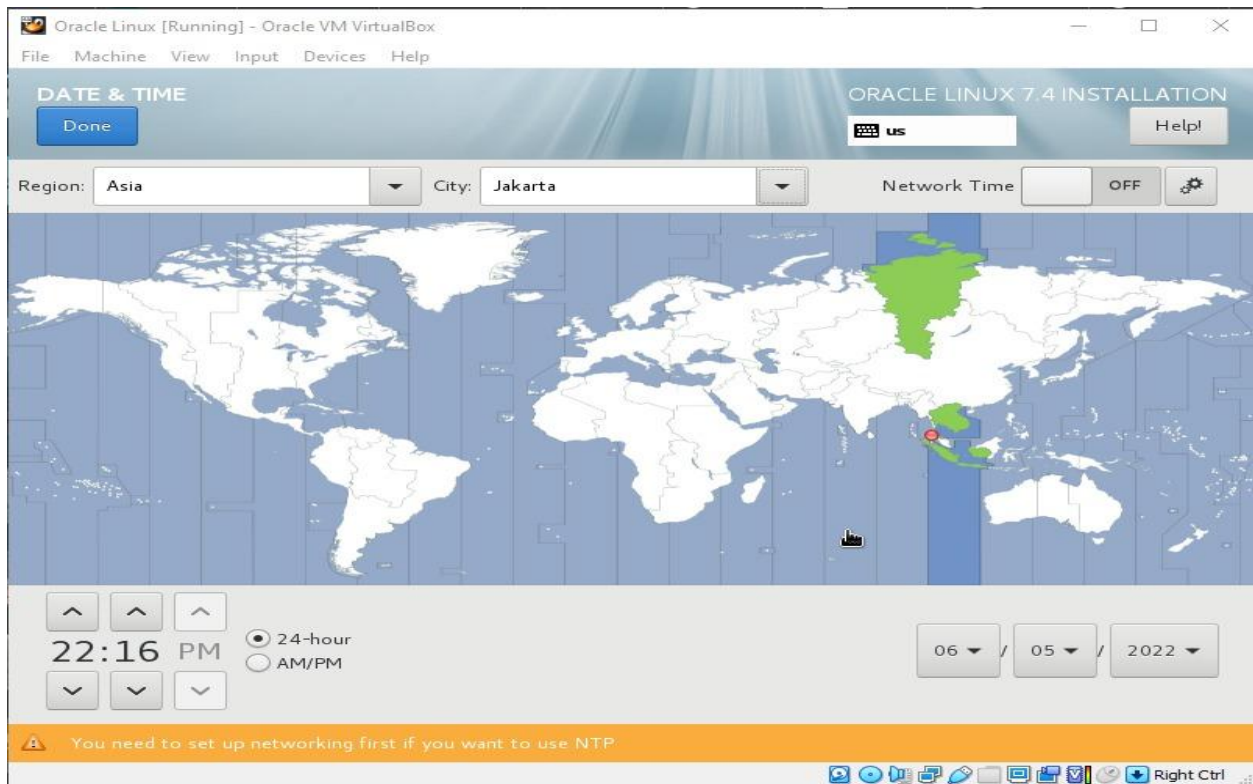
Setelah VM sudah di set dan di start, selanjutnya masuk ke tahap install Oracle Linux 7.

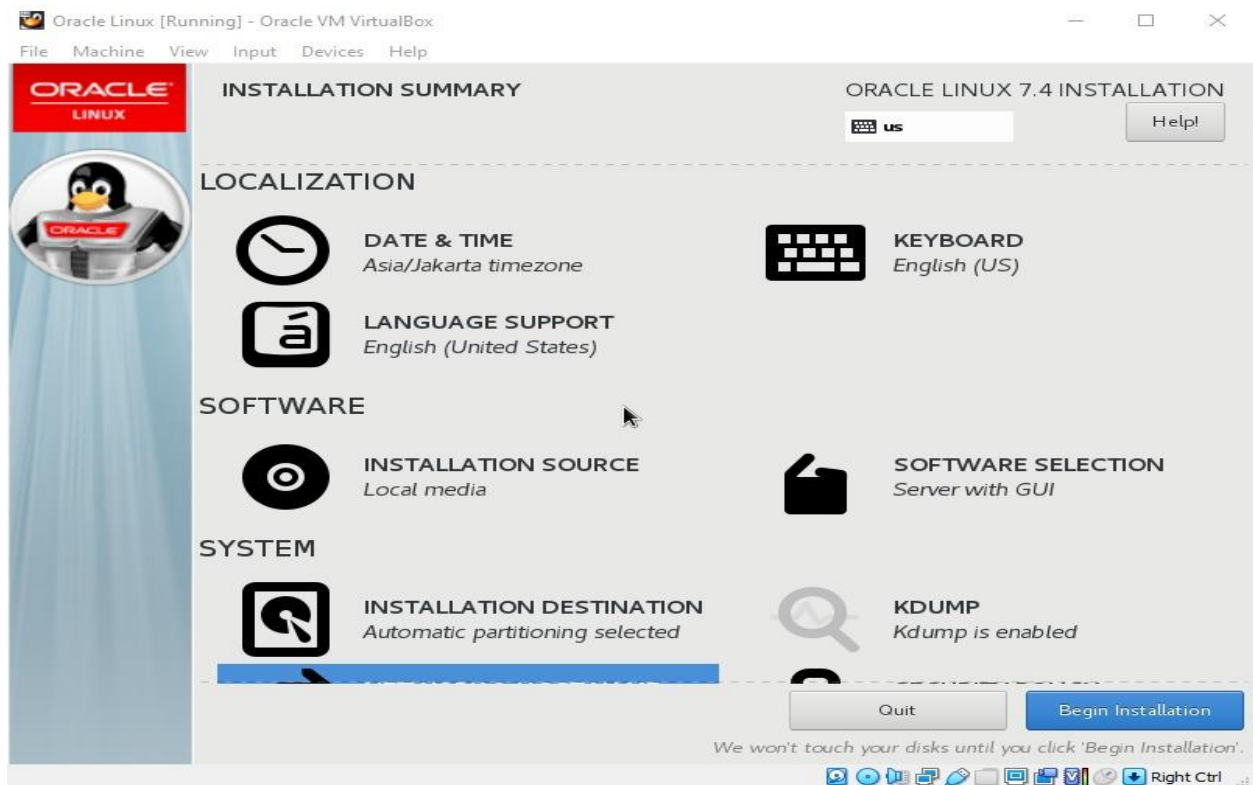


Set language untuk server Oracle Linux 7.

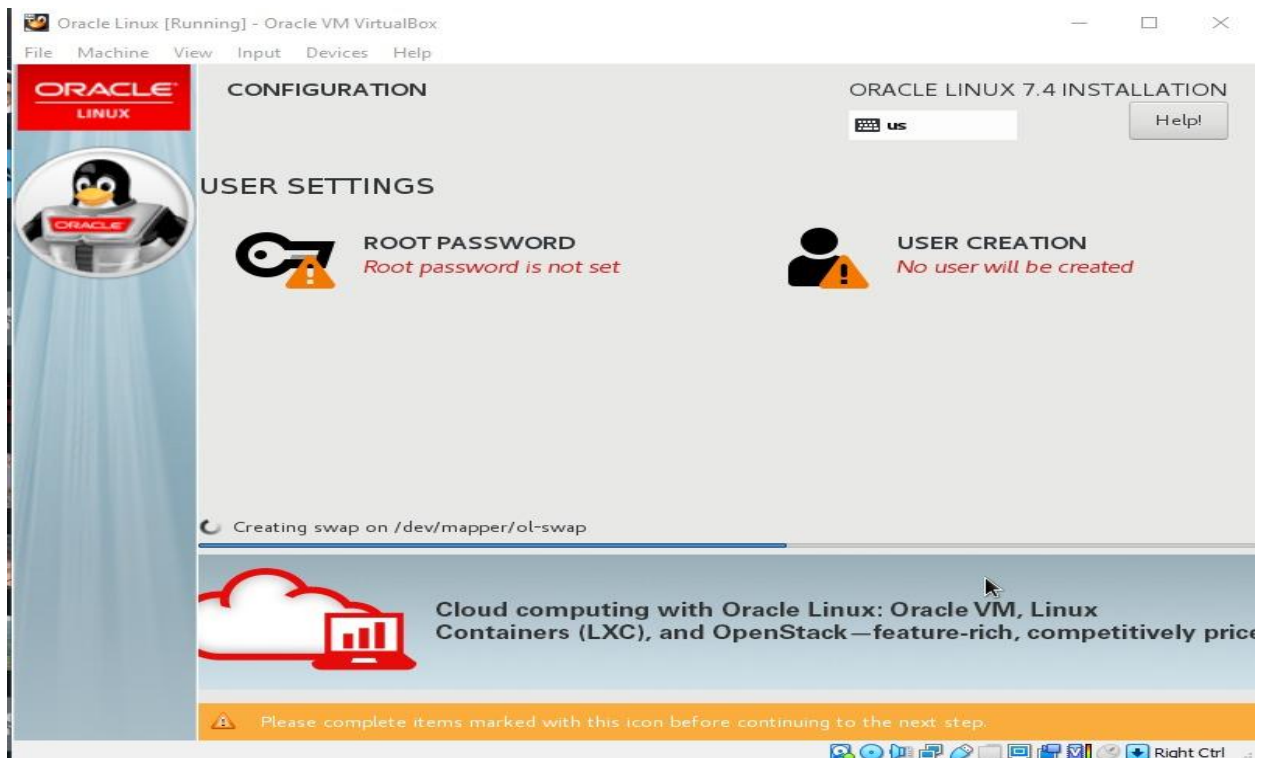


Set date & time, installation source, software selection (install with GUI), dan installation destinationnya.

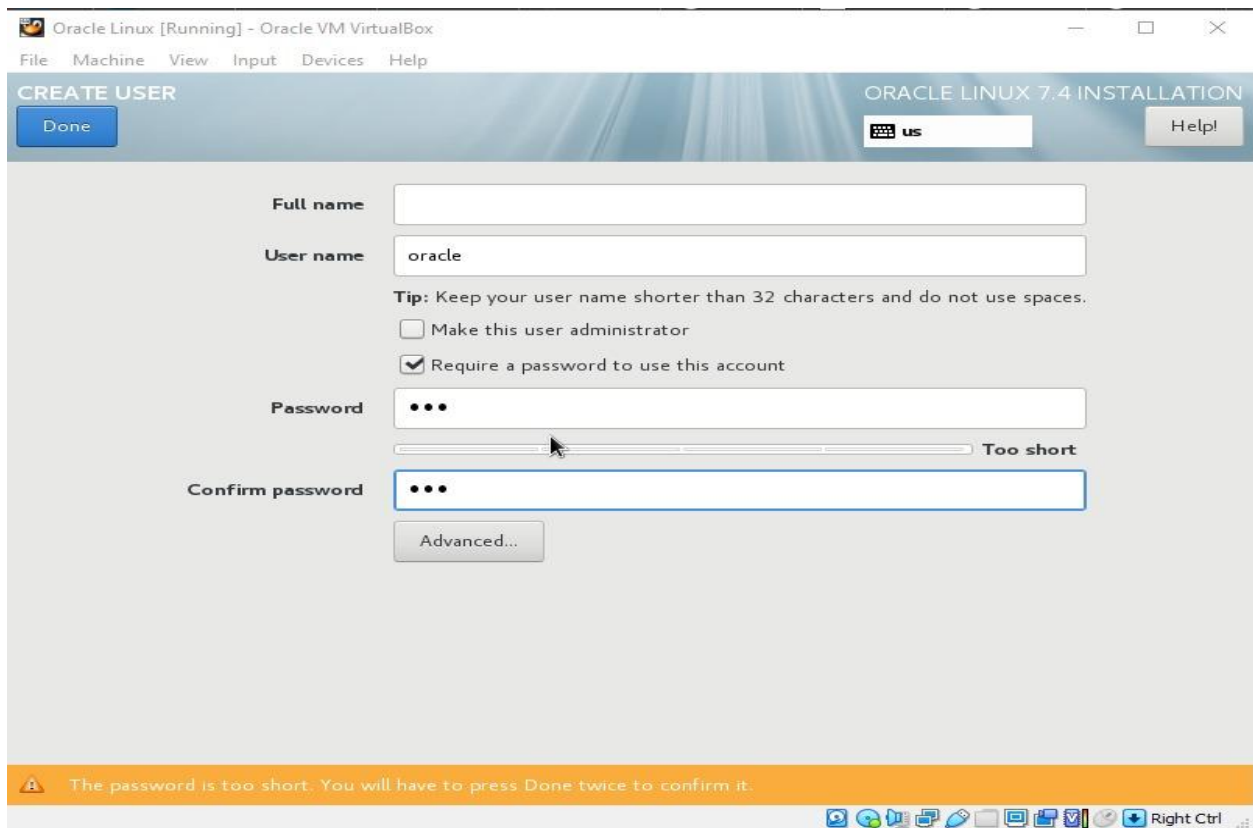
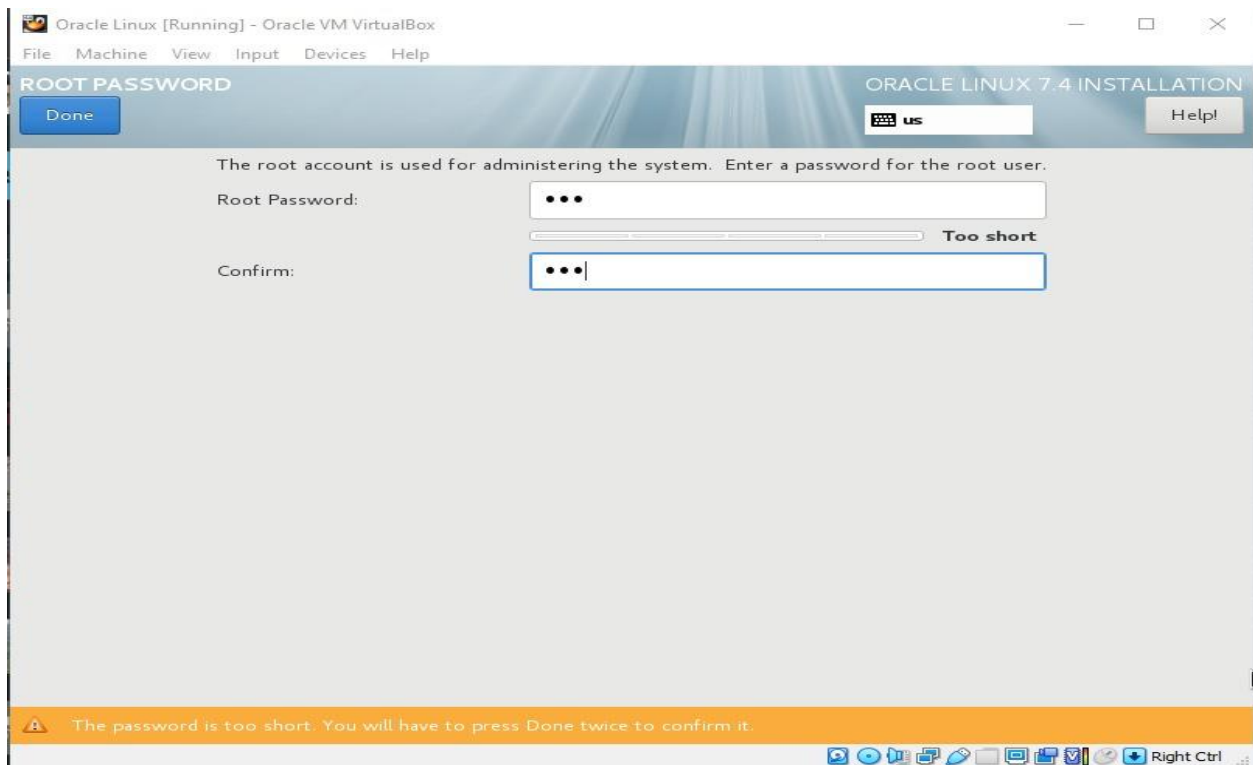




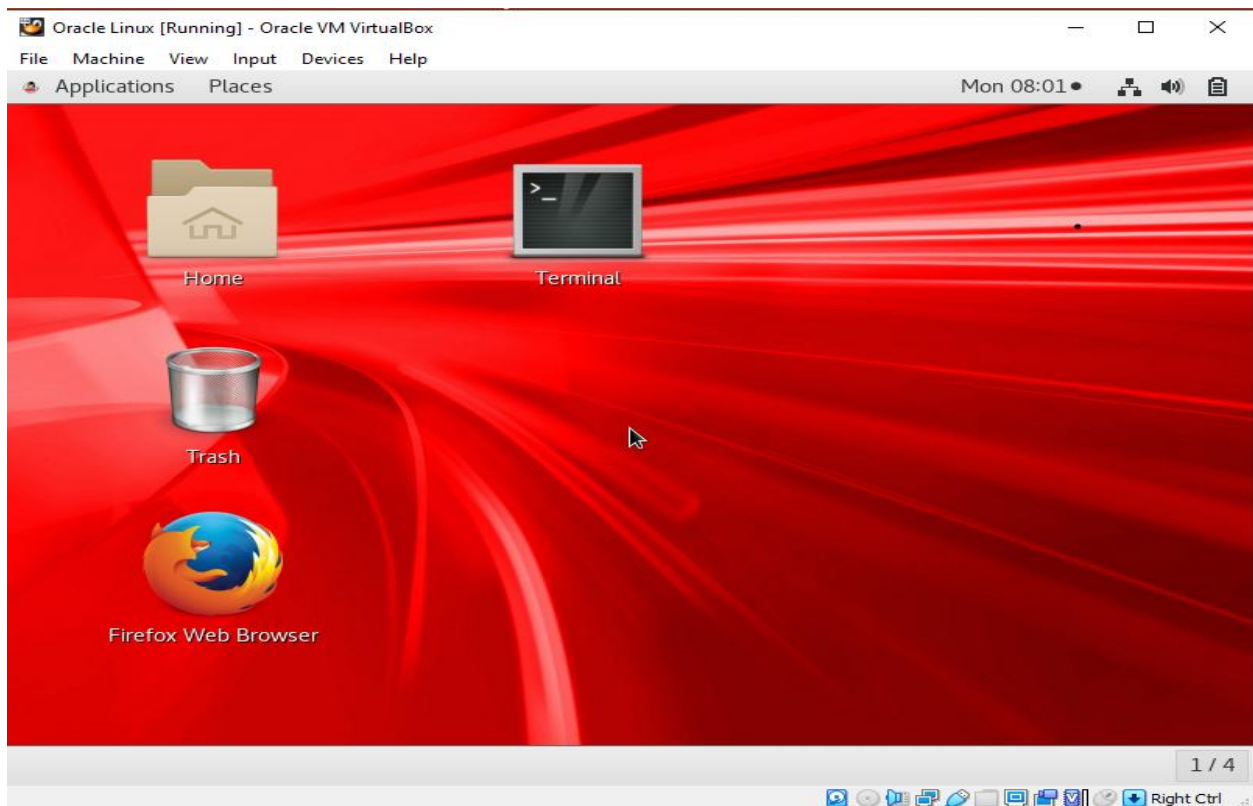
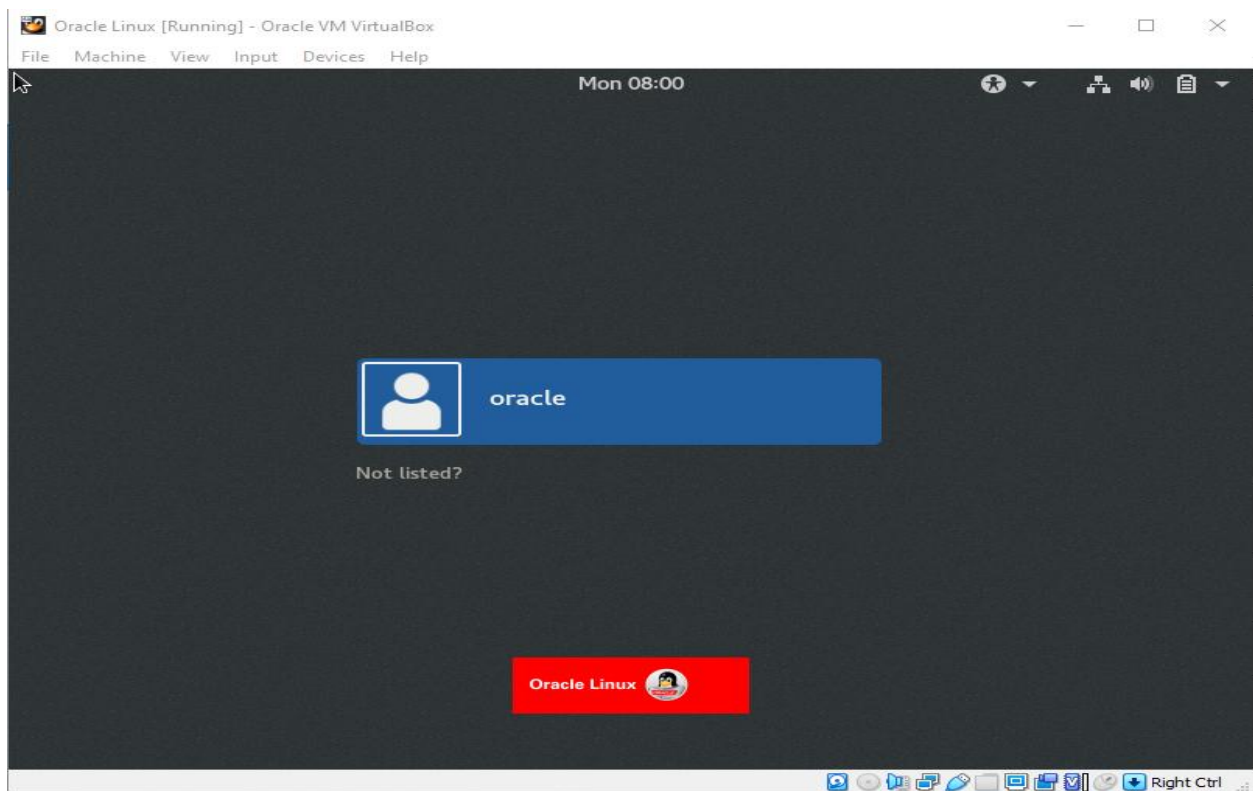
Setelah semua sudah di set, lalu step selanjutnya adalah begin installation.



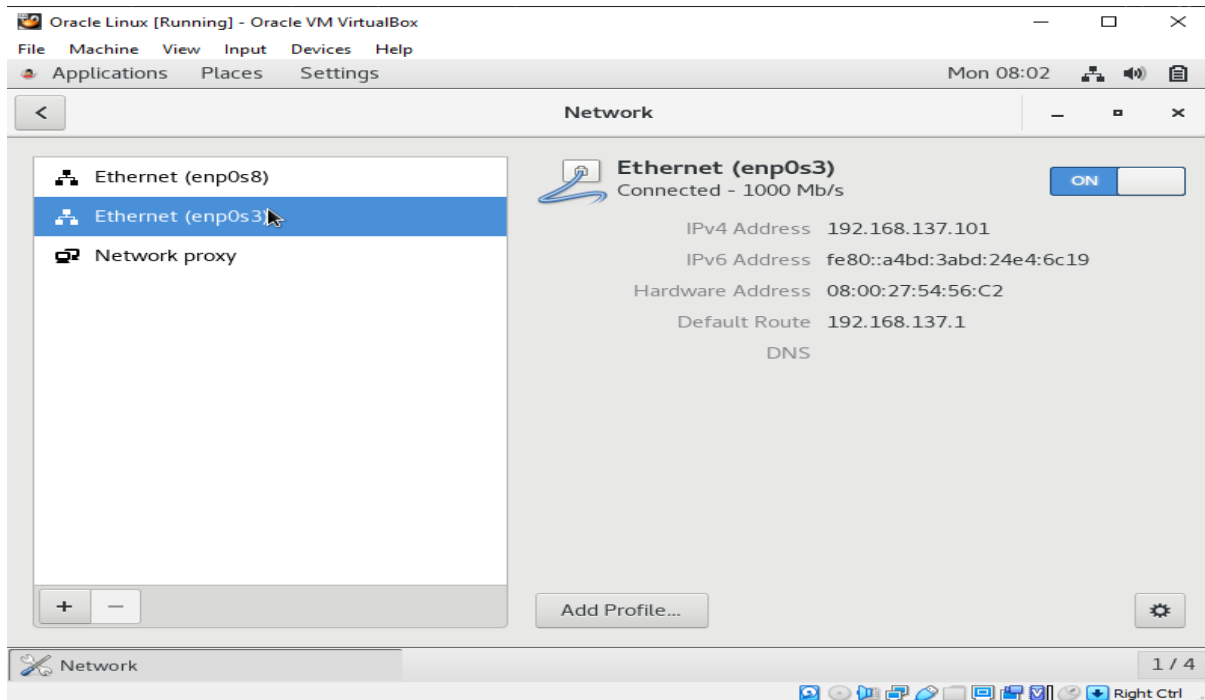
Lalu melakukan set root password dan membuat user baru.



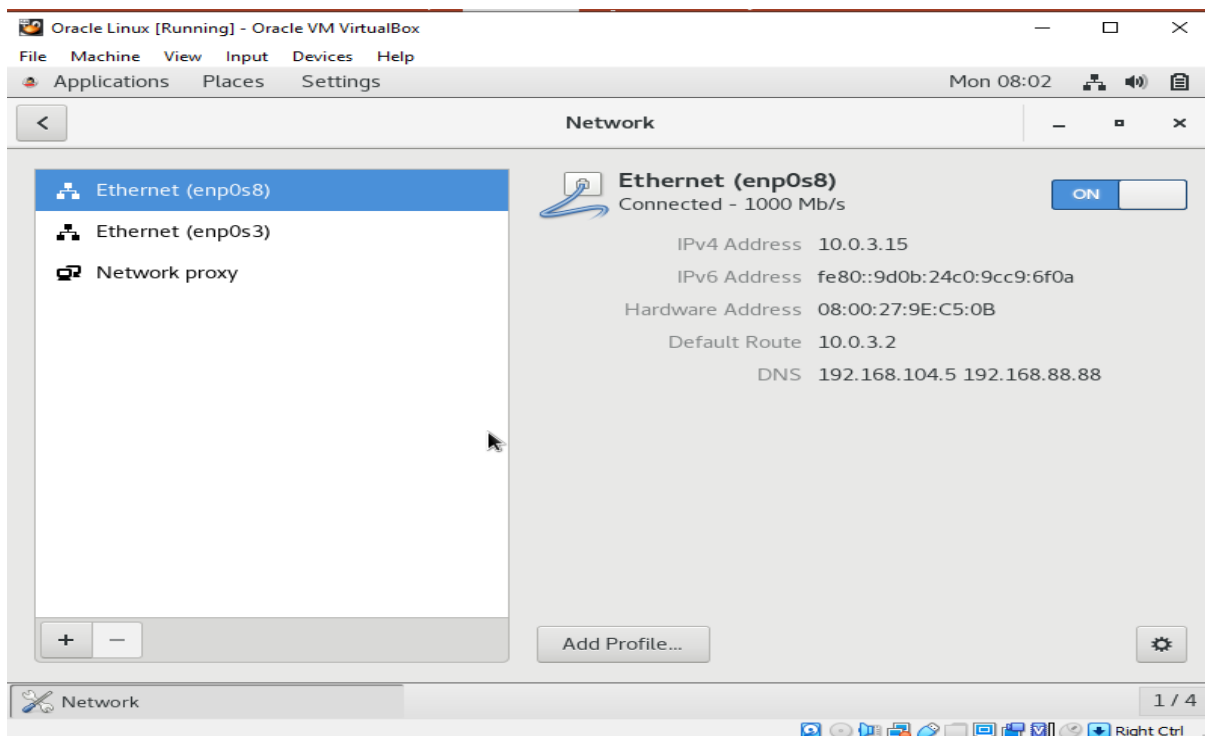
Saat semua konfigurasi sudah di set, tunggu proses instalasi selesai.



Oracle linux 7 berhasil di install.



Set ethernet (enp0s3) menjadi connect automatic.



Set ethernet (enp0s8) dengan menyesuaikan IPv4 manual, lalu set seperti gambar dibawah.

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

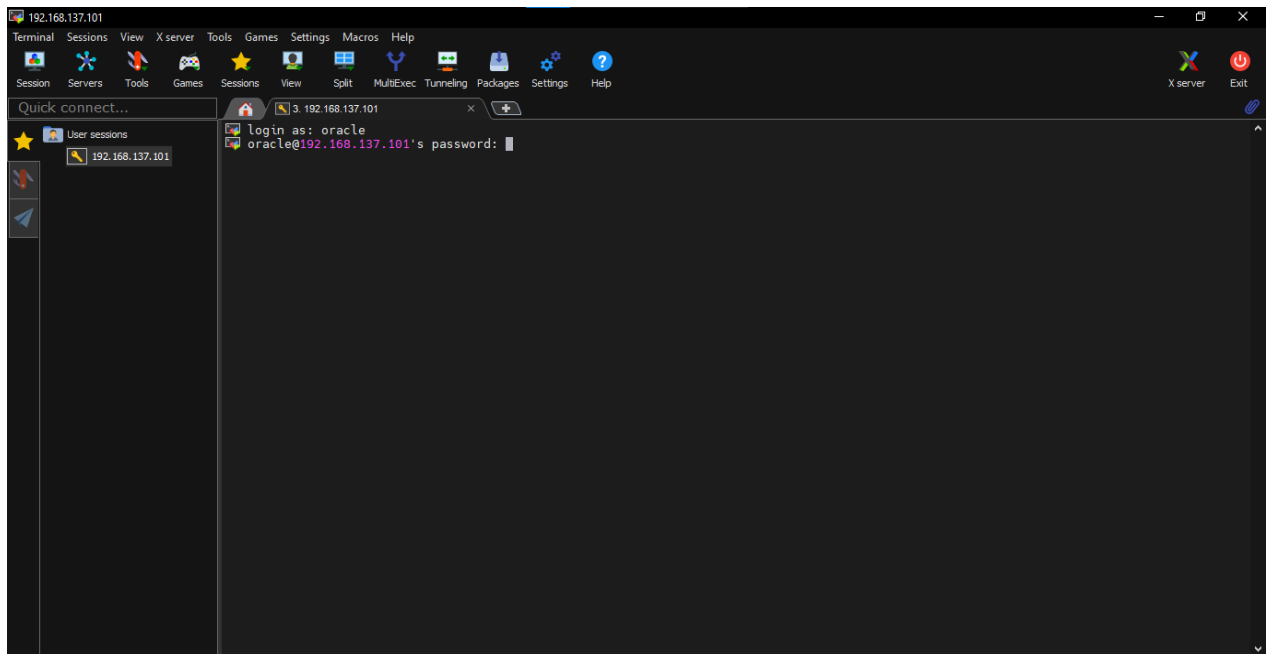
Addresses

Address	Netmask	Gateway	
192.168.56.10	255.255.255.255	192.168.56.1	✕
			✕

DNS Automatic ON

Separate IP addresses with commas

Set address, netmask, dan gateway untuk di ssh lewat MobaXterm. (address OPTIONAL).



Setelah itu, ssh server yang sudah dibuat tadi melalui MobaXterm.

Install Database 19c on VM

1. Masuk ke user root di MobaXterm

```
su – root
```

2. Edit file di /etc/hosts

```
vi /etc/hosts
```

3. Masukkan IP yang di sett pada awal tadi (addres), dan masukan hostname yang diinginkan di paling bawah bagian /etc/hosts.

```
exaprimary    192.168.56.10
```

4. Sett hostname yang kamu inginkan dengan mengedit /etc/hostname

```
vi /etc/hostname
```

5. Lalu masukan hostname yang kamu inginkan

```
exaprimary
```

6. Setelah itu lakukan reboot pada server agar hostname dan hosts yang kamu edit ter-apply.

AUTOMATIC SETUP

7. If you plan to use the "oracle-database-preinstall-19c" package to perform all your prerequisite setup, issue the following command.

```
yum install -y oracle-database-preinstall-19c
```

8. Setelah proses preinstall selesai lalu lakukan update pada server tersebut.

```
yum update -y
```

MANUAL SETUP

9. If you have not used the "oracle-database-preinstall-19c" package to perform all prerequisites, you will need to manually perform the following setup tasks.

Add the following lines to the "/etc/sysctl.conf" file, or in a file called "/etc/sysctl.d/98-oracle.conf".

```
fs.file-max = 6815744
kernel.sem = 250 32000 100 128
kernel.shmmni = 4096
kernel.shmall = 1073741824
kernel.shmmax = 4398046511104
kernel.panic_on_oops = 1
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048576
net.ipv4.conf.all.rp_filter = 2
net.ipv4.conf.default.rp_filter = 2
fs.aio-max-nr = 1048576
net.ipv4.ip_local_port_range = 9000 65500
```


10. Run one of the following commands to change the current kernel parameters, depending on which file you edited.

```
/sbin/sysctl -p
```

```
# Or
```

```
/sbin/sysctl -p /etc/sysctl.d/98-oracle.conf
```

11. Add the following lines to a file called "/etc/security/limits.d/oracle-database-preinstall-19c.conf" file.

```
oracle soft nfile 1024
oracle hard nfile 65536
oracle soft nproc 16384
oracle hard nproc 16384
oracle soft stack 10240
oracle hard stack 32768
oracle hard memlock 134217728
oracle soft memlock 134217728
```

NOTES AFTER INSTALATION.

Perihal instalasi menggunakan Automatic Setup atau manual Setup, keduanya optional, dan sesuai kebutuhan kalian. Namun kalau untuk Server Lab saja, saya merekomendasikan Automatic Setup. Karna lebih cocok untuk pemula, dan lebih mudah.

12. The following packages are listed as required. Many of the packages should be installed already.

```
yum install -y bc
yum install -y binutils
yum install -y compat-libcap1
yum install -y compat-libstdc++-33
#yum install -y dtrace-modules
#yum install -y dtrace-modules-headers
#yum install -y dtrace-modules-provider-headers
yum install -y dtrace-utils
yum install -y elfutils-libelf
yum install -y elfutils-libelf-devel
yum install -y fontconfig-devel
yum install -y glibc
yum install -y glibc-devel
yum install -y ksh
yum install -y libaio
yum install -y libaio-devel
yum install -y libdtrace-ctf-devel
yum install -y libXrender
yum install -y libXrender-devel
yum install -y libX11
yum install -y libXau
yum install -y libXi
yum install -y libXtst
yum install -y libgcc
yum install -y librdmacm-devel
yum install -y libstdc++
yum install -y libstdc++-devel
yum install -y libxcb
yum install -y make
yum install -y net-tools # Clusterware
yum install -y nfs-utils # ACFS
yum install -y python # ACFS
yum install -y python-configshell # ACFS
yum install -y python-rtlib # ACFS
yum install -y python-six # ACFS
yum install -y targetcli # ACFS
yum install -y smartmontools
yum install -y sysstat
```

```
yum install -y gcc
yum install -y unixODBC
```

13. Create the new groups and users.

```
groupadd -g 54321 oinstall
groupadd -g 54322 dba
groupadd -g 54323 oper
#groupadd -g 54324 backupdba
#groupadd -g 54325 dgdba
#groupadd -g 54326 kmdba
#groupadd -g 54327 asmdba
#groupadd -g 54328 asmoper
#groupadd -g 54329 asmadmin
#groupadd -g 54330 racdba

useradd -u 54321 -g oinstall -G dba,oper oracle
```

ADDITIONAL SETUP

The following steps must be performed, whether you did the manual or automatic setup.

14. Set the password for the "oracle" user.

```
passwd oracle
```

15. Set secure Linux to permissive by editing the "/etc/selinux/config" file, making sure the SELINUX flag is set as follows.

```
SELINUX=permissive
```

16. Once the change is complete, restart the server or run the following command.

```
setenforce Permissive
```

17. If you have the Linux firewall enabled, you will need to disable or configure it, as shown here. To disable it, do the following.

```
systemctl stop firewalld
systemctl disable firewalld
```

18. Create the directories in which the Oracle software will be installed.

```
mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
mkdir -p /u02/oradata
chown -R oracle:oinstall /u01 /u02
chmod -R 775 /u01 /u02
```

19. Create a "scripts" directory.

```
mkdir /home/oracle/scripts
```

20. Create an environment file called "setEnv.sh". The "\$" characters are escaped using "\". If you are not creating the file with the cat command, you will need to remove the escape characters.

```
cat > /home/oracle/scripts/setEnv.sh <<EOF
# Oracle Settings
export TMP=/tmp
export TMPDIR=\$TMP

export ORACLE_HOSTNAME=ol7-19.localdomain
export ORACLE_UNQNAME=cdb1
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=\$ORACLE_BASE/product/19.0.0/dbhome_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=cdb1
export PDB_NAME=pdb1
export DATA_DIR=/u02/oradata

export PATH=/usr/sbin:/usr/local/bin:\$PATH
export PATH=\$ORACLE_HOME/bin:\$PATH

export LD_LIBRARY_PATH=\$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=\$ORACLE_HOME/jlib:\$ORACLE_HOME/rdbms/jlib
EOF
```

21. Add a reference to the "setEnv.sh" file at the end of the "/home/oracle/.bash_profile" file.

```
echo ". /home/oracle/scripts/setEnv.sh" >> /home/oracle/.bash_profile
```

22. Create a "start_all.sh" and "stop_all.sh" script that can be called from a startup/shutdown service. Make sure the ownership and permissions are correct.

```
cat > /home/oracle/scripts/start_all.sh <<EOF
#!/bin/bash
. /home/oracle/scripts/setEnv.sh

export ORAENV_ASK=NO
. oraenv
export ORAENV_ASK=YES

dbstart \${ORACLE_HOME}
EOF
```

```
cat > /home/oracle/scripts/stop_all.sh <<EOF
#!/bin/bash
. /home/oracle/scripts/setEnv.sh

export ORAENV_ASK=NO
. oraenv
export ORAENV_ASK=YES

dbshut \${ORACLE_HOME}
EOF
```

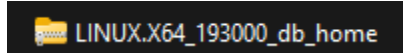
```
chown -R oracle:oinstall /home/oracle/scripts
chmod u+x /home/oracle/scripts/*.sh
```

Installation Oracle 19c

1. Masuk ke directories \$ORACLE_HOME

```
/u01/app/oracle/product/19.0.0/dbhome_1
```

2. Upload database home 19c pada directories \$ORACLE_HOME.

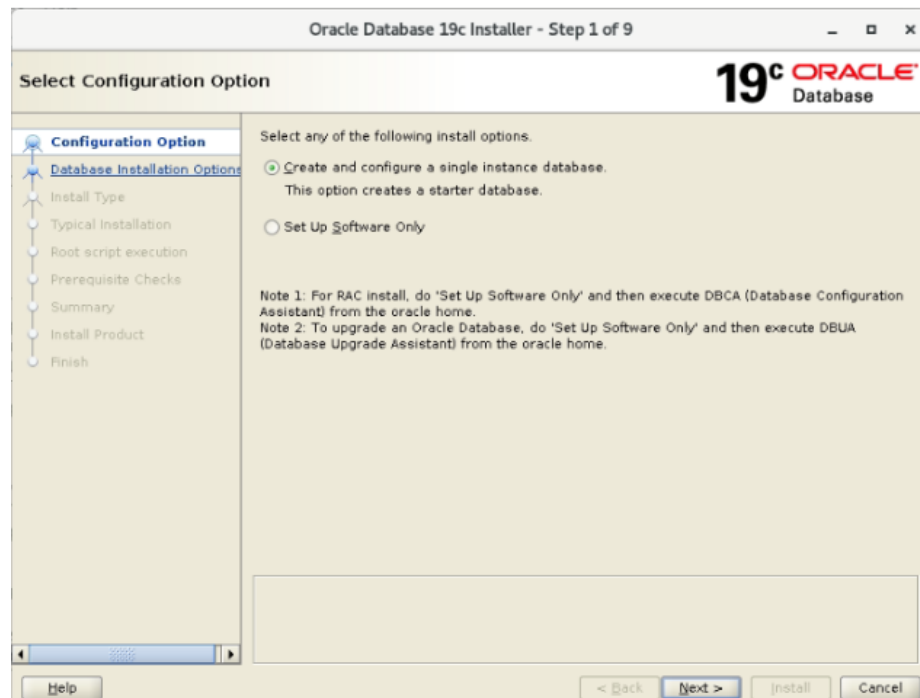
A terminal window with a dark background. The prompt is a yellow icon followed by a root symbol. The command 'LINUX.X64_193000_db_home' is entered and highlighted in blue.

3. Setelah sudah di upload lakukan unzip.

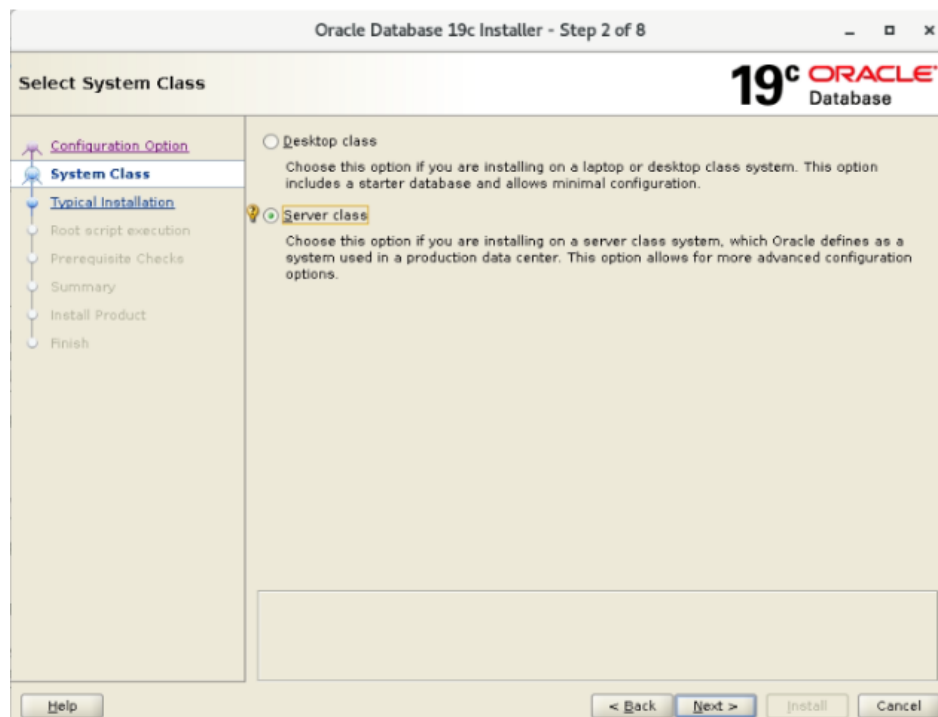
```
unzip -oq LINUX.X64_19.3000_db_home
```

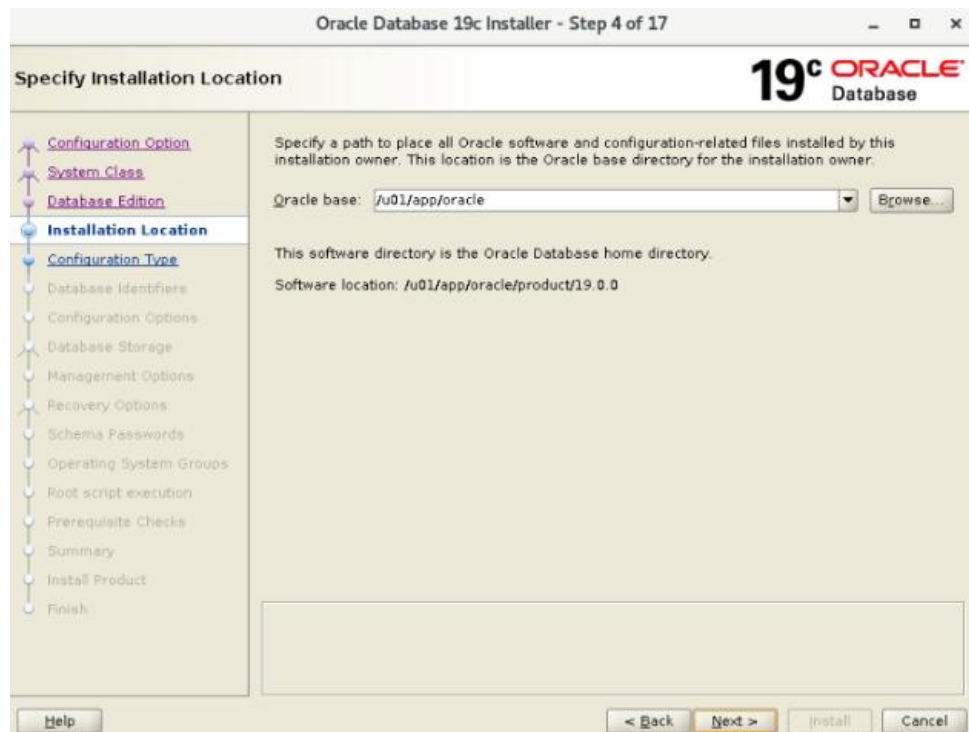
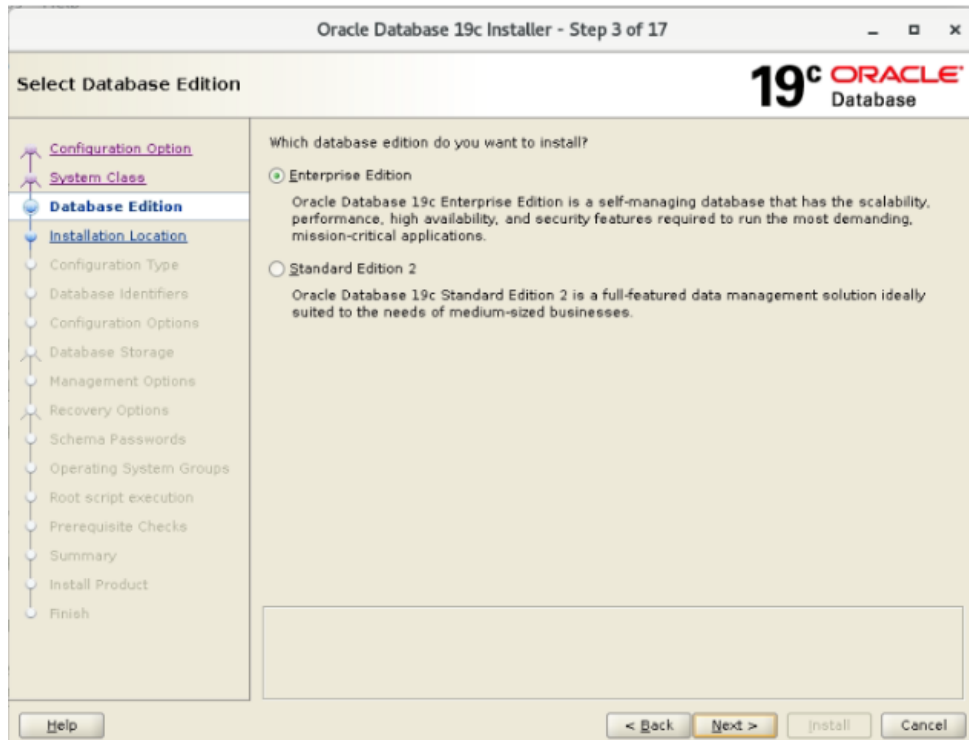
4. Setelah proses unzip selesai jalankan runinstaller yang ada.

```
./runInstaller
```

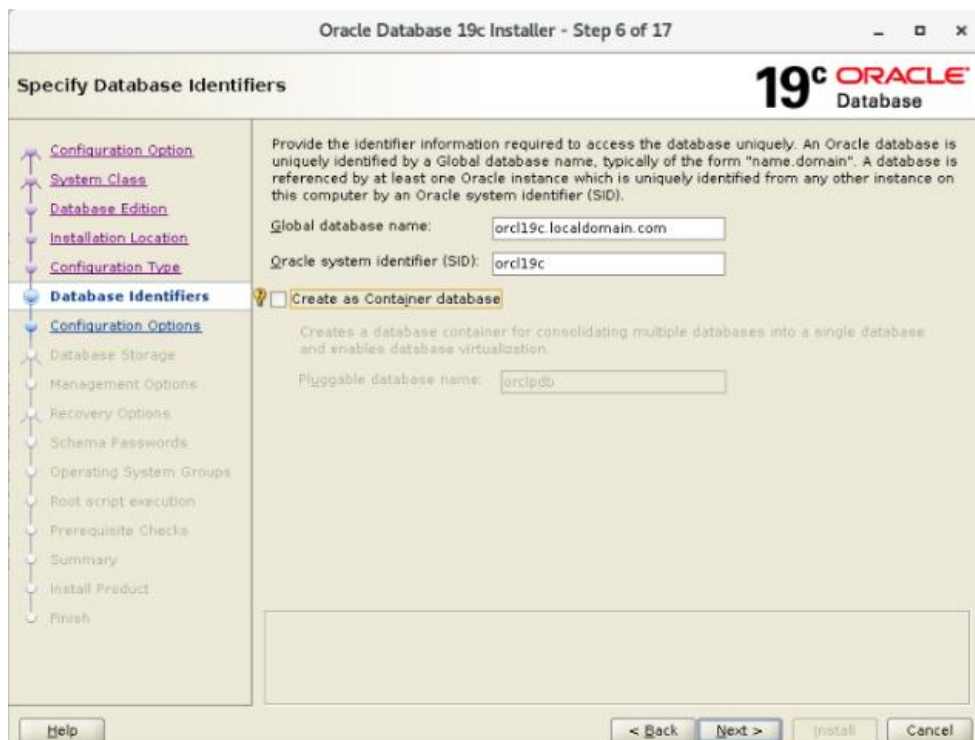
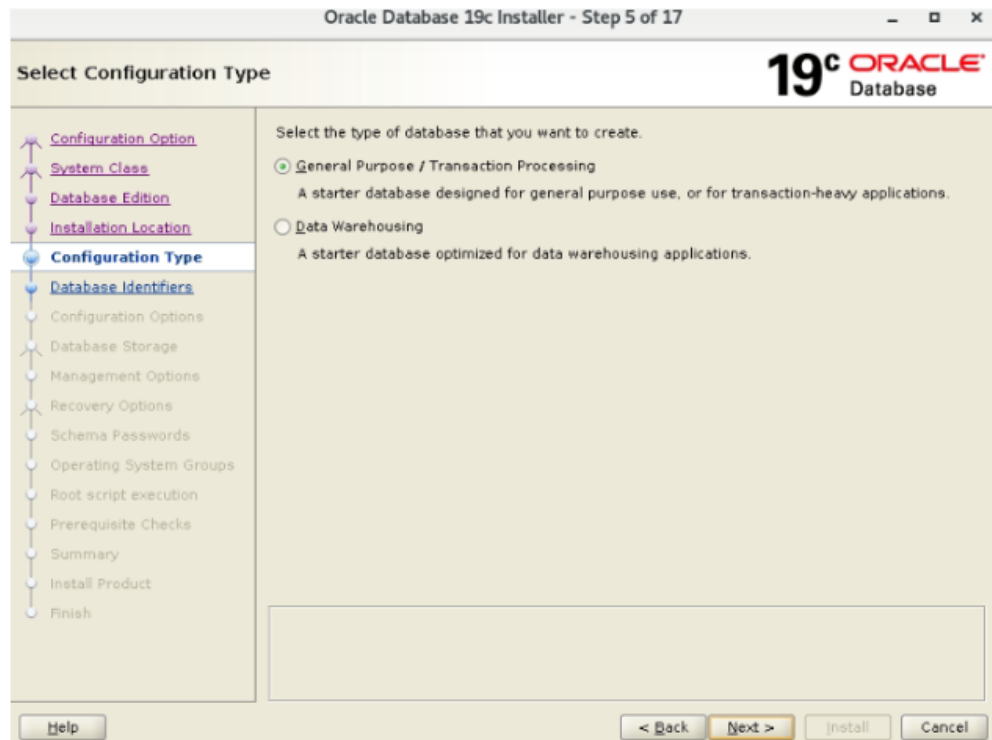


Pilih sesuai kebutuhan, create and configure a single instance database, untuk menginstall software sekaligus create database, dan set up software only hanya untuk menginstall software oracle database 19c.

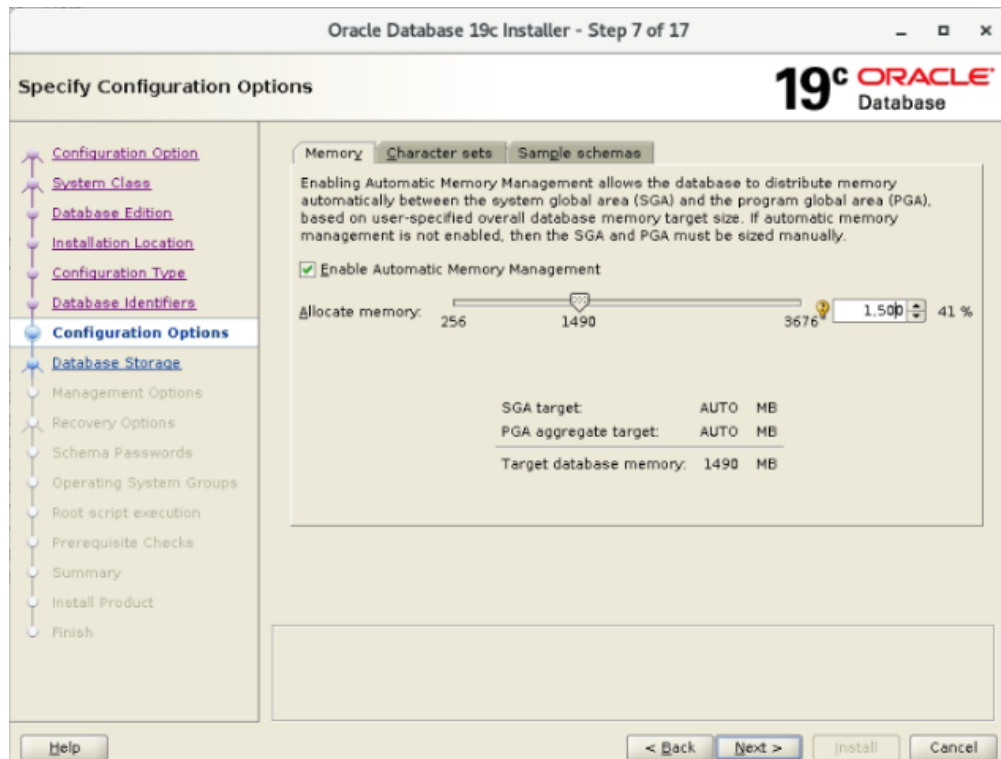




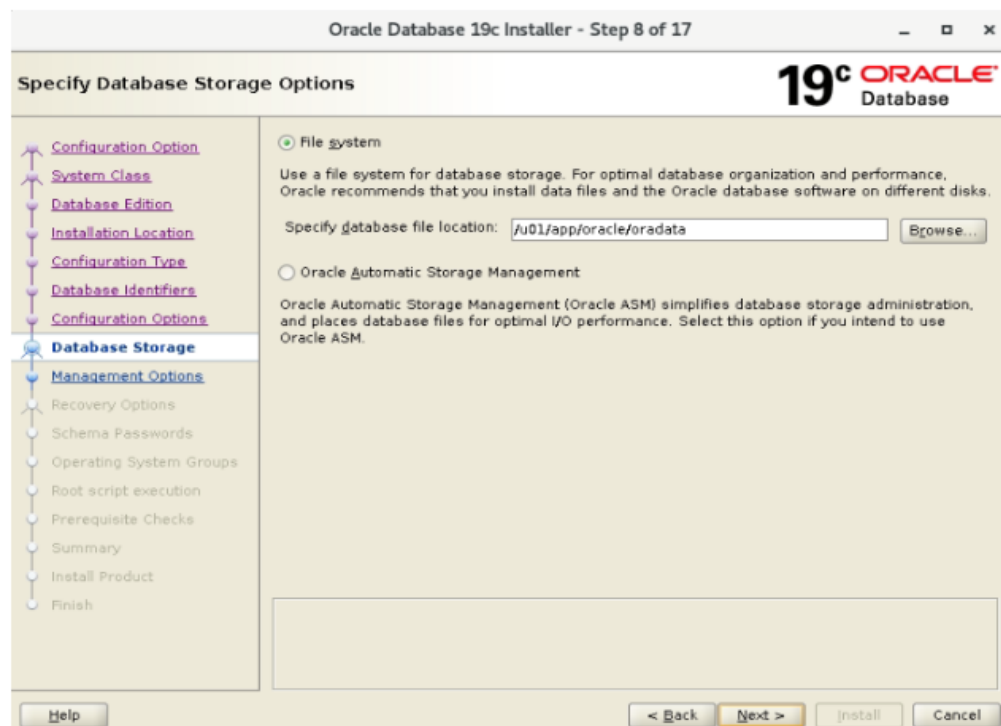
Set oracle base sesuai kebutuhan. (OPTIONAL)



Sesuaikan database name dan SID sesuai kebutuhan. (OPTIONAL)



Set Memory SGA & PGA. (default recommendation for lab)



Set path spesifik untuk database file.

Oracle Database 19c Installer - Step 9 of 17

Specify Management Options

19c ORACLE Database

Oracle Database 12c is managed by Oracle Database Express by default. You can use Oracle Enterprise Manager 12c Cloud Control to manage each Oracle Database 19c centrally. Specify the details of the Cloud Control configuration to manage your database.

☐ Register with Enterprise Manager (EM) Cloud Control

OMS Host:

OMS Port:

EM Admin User Name:

EM Admin Password:

Management Options

- Configuration Option
- System Class
- Database Edition
- Installation Location
- Configuration Type
- Database Identifiers
- Configuration Options
- Database Storage
- Management Options**
- Recovery Options
- Schema Passwords
- Operating System Groups
- Root script execution
- Prerequisite Checks
- Summary
- Install Product
- Finish

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 10 of 17

Specify Recovery Options

19c ORACLE Database

Enable or disable recovery for your database. If you choose to enable recovery, then the location specified will be used as recovery area storage.

☒ **Enable Recovery**

☒ File system

Recovery area location: Browse...

☐ Oracle Automatic Storage Management

Recovery Options

- Configuration Option
- System Class
- Database Edition
- Installation Location
- Configuration Type
- Database Identifiers
- Configuration Options
- Database Storage
- Management Options
- Recovery Options**
- Schema Passwords
- Operating System Groups
- Root script execution
- Prerequisite Checks
- Summary
- Install Product
- Finish

Help < Back Next > Install Cancel

Set recovery, perihal location OPTIONAL.

Oracle Database 19c Installer - Step 11 of 17

Specify Schema Passwords

The starter database contains pre-loaded schemas, most of which have passwords that are expired and locked at the end of installation. After installation is complete, you must unlock and set new passwords for those accounts you want to use. Schemas used for database management and postinstallation functions are left unlocked, and passwords for these accounts will not expire. Specify the passwords for these accounts.

☐ Use different passwords for these accounts

	Password	Confirm password
SYS		
SYSTEM		

☒ Use the same password for all accounts

Password: Confirm password:

Messages:

 Password: [INS-30011] The ADMIN password entered does not conform to the Oracle recommended standards.

Help < Back Next > Install Cancel

Sett password semua account sama atau berbeda. (OPTIONAL)

Oracle Database 19c Installer - Step 12 of 17

Privileged Operating System groups

SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.

Database Administrator (OSDBA) group: dba

Database Operator (OSOPER) group (Optional): oinstall

Database Backup and Recovery (OSBACKUPDBA) group: oinstall

Data Guard administrative (OSDGDBA) group: oinstall

Encryption Key Management administrative (OSKMDBA) group: oinstall

Real Application Cluster administrative (OSRACDBA) group: oinstall

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 13 of 17

Root script execution configuration

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

☐ Use sudo

Program path :

User name :

Password :

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 14 of 17

Perform Prerequisite Checks

Verification Result

All minimum requirements are satisfied. You may proceed with the installation.

☐ Ignore All

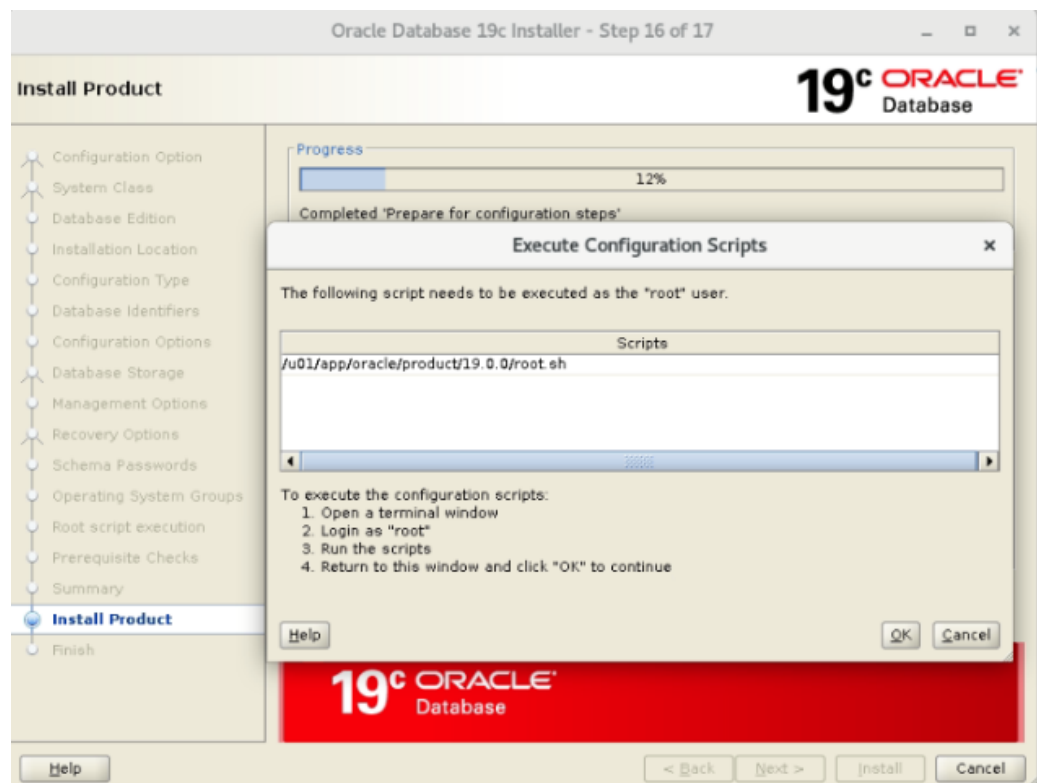
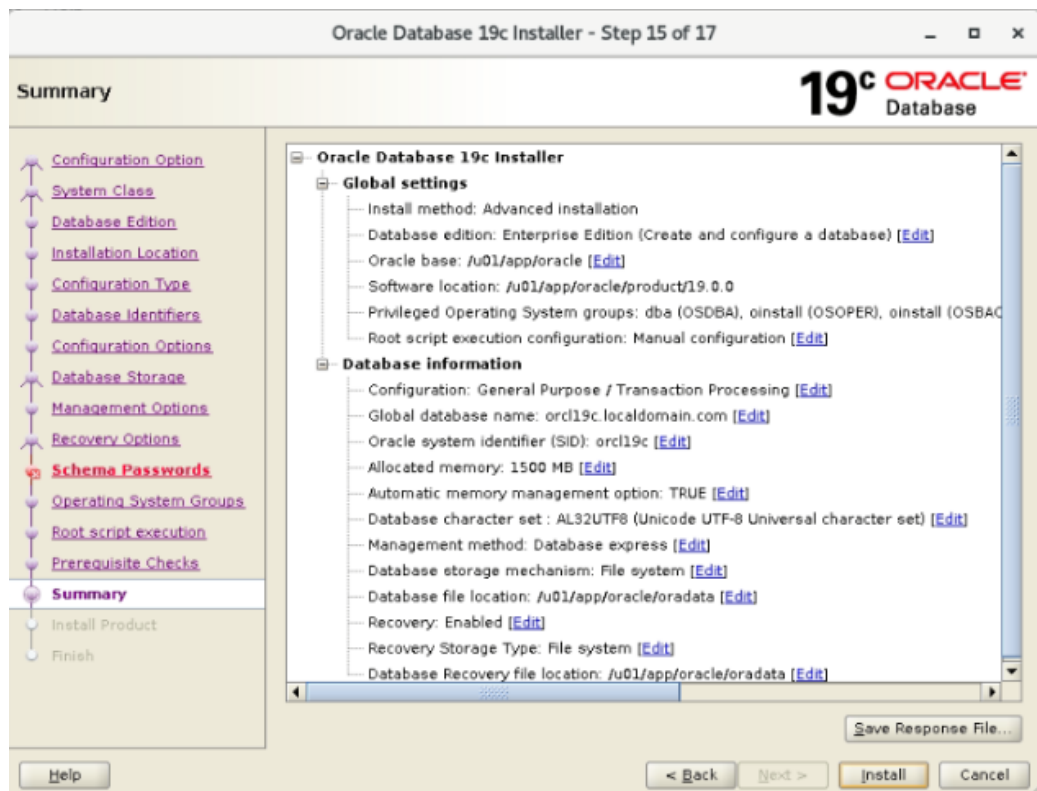
Checks	Status	Fixable
Physical Memory	Succeeded	
Available Physical Memory	Succeeded	
Swap Size	Succeeded	
Free Space	Succeeded	
Free Space: 19c/tmp	Succeeded	
User Existence	Succeeded	
User Existence: oracle	Succeeded	
Users With Same UID: 54321	Succeeded	
Group Existence	Succeeded	
Group Existence: dba	Succeeded	
Group Existence: oinstall	Succeeded	
Group Membership	Succeeded	
Group Membership: oinstall	Succeeded	
Group Membership: dba	Succeeded	
Run Level	Succeeded	
Architecture	Succeeded	

This is a prerequisite condition to test whether the system has at least 1GB (1048576.0KB) of total physical memory: [more details](#)

Expected Value : 1GB (1048576.0KB)
Actual Value : 3.5914GB (3765840.0KB)

Help < Back Next > Install Cancel

Jika ada alert swap, bisa di ignore all semisal untuk lab.



Jalankan script dengan user ROOT.

Finish

- Configuration Option
- System Class
- Database Edition
- Installation Location
- Configuration Type
- Database Identifiers
- Configuration Options
- Database Storage
- Management Options
- Recovery Options
- Schema Passwords
- Operating System Groups
- Root script execution
- Prerequisite Checks
- Summary
- Install Product

Finish

The configuration of Oracle Database was successful.

Note:

Oracle Enterprise Manager Database Express URL: <https://fram.doyen.com:5500/em>

[Help](#)

< Back

Next >

Install

Close

Startup Database

1. Remote / ssh ke server dengan user oracle
Remote menggunakan moba/putty ke IP servernya
2. Jalankn profile database yang ada
.`./act`

sett (pilih database)

untuk memasstikan :

echo \$ORACLE_HOME
3. Login ke SQL*Plus
\$> sqlplus / as sysdba (login ke sqlplus sebagai user sys)
4. Sql> startup

Shutdown Database

1. Remote / ssh ke server dengan user oracle
Remote menggunakan moba/putty ke IP servernya
2. Jalankn profile database yang ada
.`./act`

sett (pilih database)
3. Login ke SQL*Plus
\$> sqlplus / as sysdba (login ke sqlplus sebagai user sys)
4. Check db name untuk memastikan di database yang benar
SQL> select name, database_role, open_mode from v\$database;
5. SQL> shutdown immediate

Check Status & start/stop Listener

1. Remote / ssh ke server dengan user oracle
Remote menggunakan moba/putty ke IP servernya

2. Jalankn profile database yang ada :
./act

set (pilih database)

cd \$ORACLE_HOME/network/admin

Vi listener.ora // cari nama listener yang mau di cek

3. ===Check status listener===
lsnrctl status (nama listener)

===start listener===

lsnrctl start

===stop listener===

lsnrctl stop

CHECK ADA ATAU TIDAKNYA DATABASE DI SERVER

1. ketik command `ps -ef | grep pmon` (jika database ada dan hidup maka akan keluar seperti gambar dibawah)

```
[oracle@UBPBDBDC ~]$ ps -ef | grep pmon
oracle      845735      1  0  2023 ?        00:21:58 ora_pmon_UBPCORE
oracle     3060255 3060165  0 08:59 pts/0    00:00:00 grep --color=auto pmon
```

2. atau jika database dalam keadaan mati bisa dengan cat `/etc/oratab`

```
#
UBPCORE:/u01/app/oracle/product/19.0.0/dbhome_1:N
```

Check bagaimana login dengan tepat sesuai user yang milik database nya

1. `ps -ef | grep pmon`

```
oracle      845735      1  0  2023 ?        00:21:58 ora_pmon_UBPCORE
oracle     3060255 3060165  0 08:59 pts/0    00:00:00 grep --color=auto pmon
```

2. berarti bisa kita login dengan user oracle, di sisi lain jika database ownershipnya adalah milik user ora11g atau ora12c maka kita harus login server dengan user tersebut.

Check size tablespace

1. Remote / ssh ke server dengan user oracle
Remote menggunakan moba/putty ke IP servernya
2. Jalankan profile database yang ada :
./act

set (pilih database)
3. Ketik : tps
\$ tps.

```
[oracle@RMT00LDB01 bin]$ tps
```

STATUS	TABLESPACE_NAME	TYPE	EXTENT_MGT	TS_size(MB)	USED	FREE	PCT_USED
ONLINE	SYSTEM	PERMANENT	LOCAL	910	910	0	99
ONLINE	SYSAUX	PERMANENT	LOCAL	690	651	39	94
ONLINE	USERS	PERMANENT	LOCAL	5	3	2	53
ONLINE	TEMP	TEMPORARY	LOCAL	32	5	27	15
ONLINE	UNDOTBS1	UNDO	LOCAL	345	0	345	0
sum				1,982	1,568	414	

Convert Physical Standby to Snapshot Standby Database

SQL> select name, database_role, open_mode from v\$database;

SQL> alter database recover managed standby database cancel;

SQL> alter database convert to snapshot standby;

SQL> alter database open;

SQL> select name, database_role, open_mode from v\$database;

=====

Convert Snapshot Standby Database to Physical Standby (from standby)

```
SQL> select name, database_role, open_mode from v$database;

SQL> shut immediate;

SQL> startup mount;

SQL> alter database convert to physical standby;

SQL> shut immediate;

SQL> startup mount;

SQL> alter database open read only;

SQL> alter database recover managed standby database disconnect from session;
```

Convert Snapshot Standby Database to Physical Standby without Open Read Only (from standby)

```
SQL> select name, database_role, open_mode from v$database;

SQL> shut immediate;

SQL> startup mount;

SQL> alter database convert to physical standby;

SQL> shut immediate;

SQL> startup mount;

SQL> alter database recover managed standby database using current logfile disconnect;
```

CHANGE OPENMODE MOUNTED TO READ OPEN ONLY

--- matikan mrp ---

```
SQL> alter database recover managed standby database cancel;
```

--- menaikan ke status open read only ---

```
SQL> alter database open read only;
```

--- menyalakan MRP ---

```
SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT  
LOGFILE DISCONNECT;
```

Add redolog group

```
SQL> ALTER DATABASE ADD LOGFILE [Group number] ('location/namafile') SIZE 1024M;
```

```
SQL> ALTER DATABASE ADD LOGFILE GROUP 4 ('/oralog/gfxdb/redo04a.log') SIZE 1024M;
```

BACKUP ORACLE DBdire

Backup DB menggunakan Schema Data Pump

1. Masuk ke profil act . ./act
2. sett ke db yang ingin di backup

Export Table

```
expdp \"/ as sysdba\" TABLES=EPURSEUSER.ACK_RECON_FILE  
dumpfile=ACK_RECON_FILE_%U.dmp logfile=ACK_RECON_FILE.log directory=DUMP;
```

Export Schemas

```
expdp \"/ as sysdba\" SCHEMAS=EPURSEUSER dumpfile=EPURSEUSER_%U.dmp  
logfile=EPURSEUSER.log directory=DUMP
```

CREATE TABLESPACE

```
SQL> create tablespace FSCM datafile '/oradata/FSCMDB/FSCM01.dbf' size 1G autoextend on next 100m;
```

CREATE USER

```
SQL> create user FSCMUSER identified by FSCMUSER123 default tablespace FSCM;
```

```
SQL> grant unlimited tablespace to FSCMUSER;
```

```
SQL> grant connect,resource to FSCMUSER;
```

CREATE USER & TABLESPACE

```
SQL>  CREATE TABLESPACE MAVERICK  DATAFILE  
      '/oradata/MAVSOADV/MAVSOADV1.dbf' SIZE 10M AUTOEXTEND ON NEXT 100M  
      MAXSIZE UNLIMITED  
  
      LOGGING  
  
      ONLINE  
  
      EXTENT MANAGEMENT LOCAL AUTOALLOCATE  
  
      BLOCKSIZE 8K  
  
      SEGMENT SPACE MANAGEMENT AUTO  
  
      FLASHBACK ON;
```

```
SQL> CREATE USER MAV_APP IDENTIFIED BY mav_app123s DEFAULT TABLESPACE MAVERICK  
TEMPORARY TABLESPACE TEMP PROFILE DEFAULT ACCOUNT UNLOCK;
```

```
SQL> GRANT CONNECT, RESOURCE TO MAV_APP;
```

```
SQL> GRANT UNLIMITED TABLESPACE TO MAV_APP;
```

```
      # ALTER USER MAV_APP DEFAULT ROLE ALL;
```

Check user lock/nolock

SQL> select username, profile from dba_users where username like '[NAMA USERNYA]' order by username;

atau

SQL> select username, account_status from dba_users where username = '[NAMA USER]';

Unlock User

SQL> alter user namauser account unlock;

SQL> alter user dbsnmp account unlock;

Unlock User and Change Password

SQL> ALTER USER username IDENTIFIED BY password ACCOUNT UNLOCK;

SQL> ALTER USER dbsnmp IDENTIFIED BY dbsnmp ACCOUNT UNLOCK;

Query list PROFILES database profile

col profile for a30

col limit for a30

set lines 200

SELECT * FROM DBA_PROFILES ;

Query check profile

```
SQL> select profile, resource_name, resource_type, limit from dba_profiles where profile = 'DEFAULT'
order by resource_type, resource_name;
```

Query ganti profile

```
SQL> ALTER PROFILE '[NAMA PROFILE NYA]' LIMIT PASSWORD_LOCK_TIME UNLIMITED;
```

QUERY CHECK SIZE DATABASE

=====

```
SQL> select DF.TOTAL/1048576666 "DataFile Size Gb",
LOG.TOTAL/1048576666 "Redo Log Size Gb",
CONTROL.TOTAL/1048576 "Control File Size Mb",
(DF.TOTAL + LOG.TOTAL + CONTROL.TOTAL)/1048576666 "Total Size Gb" from dual,
(select sum(a.bytes) TOTAL from dba_data_files a) DF,
(select sum(b.bytes) TOTAL from v$log b) LOG,
(select sum((cffsz+1)*cfbsz) TOTAL from x$kcccf c) CONTROL;
```


Check SQL_TEXT Blocking Session

SQL>

col program for a20

col machine for a20

col osuser for a20

col sql_text for a50

set lines 200

```
SELECT s.blocking_session, s.module, s.program, s.machine, s.osuser, sql.sql_text
FROM v$session s, v$sqlarea sql
WHERE blocking_session is not NULL
AND s.sql_id = sql.sql_id
ORDER BY blocking_session;
```

Check Dataguard

Primary: SQL> select thread#, max(sequence#) "Last Primary Seq Generated"
from v\$archived_log val, v\$database vdb
where val.resetlogs_change# = vdb.resetlogs_change#
group by thread# order by 1;

Standby: SQL>select thread#, max(sequence#) "Last Standby Seq Applied"
from v\$archived_log val, v\$database vdb
where val.resetlogs_change# = vdb.resetlogs_change#
and val.applied in ('YES','IN-MEMORY')
group by thread# order by 1;

CHECK GAP SEQUENCE

```
SQL> alter session set nls_date_format='DD-MM-YYYY HH24:MI:SS';
```

```
SELECT A.THREAD#, B.LAST_SEQ, A.APPLIED_SEQ, A.LAST_APP_TIMESTAMP, B.LAST_SEQ-A.APPLIED_SEQ  
ARC_DIFF
```

```
FROM (SELECT THREAD#, MAX(SEQUENCE#) APPLIED_SEQ, MAX(NEXT_TIME) LAST_APP_TIMESTAMP
```

```
FROM GV$ARCHIVED_LOG WHERE APPLIED = 'YES' OR APPLIED='IN-MEMORY' GROUP BY THREAD#) A,
```

```
(SELECT THREAD#, MAX(SEQUENCE#) LAST_SEQ FROM GV$ARCHIVED_LOG GROUP BY THREAD#) B
```

```
WHERE A.THREAD#=B.THREAD#;
```

Check last sequence applied pada primary dan standby dari primary (sync archiveLog)

```
SELECT a.dest_id, a.thread#, b. last_seq, a.applied_seq, a. last_app_timestamp,
b.last_seq-a.applied_seq ARC_DIFF

FROM (SELECT dest_id, thread#, MAX(sequence#) applied_seq,
MAX(next_time) last_app_timestamp

FROM v$archived_log WHERE applied in ('YES','IN-MEMORY') GROUP BY
dest_id, thread#) a,

(SELECT thread#, MAX (sequence#) last_seq FROM v$archived_log GROUP BY
thread#) b

WHERE a.thread# = b.thread#;
```

Check Applying archiveLog On primary and Stby (check on STBY)

```
set lines 200

set pagesize 200

col message format a90

SELECT SEQUENCE#, CREATOR, TO_CHAR(FIRST_TIME, 'MM-DD-YYYY
HH24:MI:SS')TIME, APPLIED, DELETED, THREAD#, BLOCK_SIZE FROM
v$ARCHIVED_LOG

WHERE first_time>=sysdate-3

order by FIRST_TIME asc;
```

CEK STATUS MRP

```
SQL> select thread#, sequence#, process, status from gv$managed_standby where process like
'%MRP%';
```

```
SQL> select process, status, thread#, sequence# from v$managed_standby;
```

Check Jumlah Datafile dari tablespace secara manual tanpa OEM

```
SET PAUSE ON
```

```
SET PAUSE 'Press Return to Continue'
```

```
SET PAGESIZE 60
```

```
SET LINESIZE 300
```

```
COLUMN "Tablespace Name" FORMAT A20
```

```
COLUMN "File Name" FORMAT A80
```

```
SELECT Substr(df.tablespace_name,1,20) "Tablespace Name",
       Substr(df.file_name,1,80) "File Name",
       Round(df.bytes/1024/1024,0) "Size (M)",
       decode(e.used_bytes,NULL,0,Round(e.used_bytes/1024/1024,0)) "Used (M)",
       decode(f.free_bytes,NULL,0,Round(f.free_bytes/1024/1024,0)) "Free (M)",
       decode(e.used_bytes,NULL,0,Round((e.used_bytes/df.bytes)*100,0)) "% Used"
FROM   DBA_DATA_FILES DF,
       (SELECT file_id,
              sum(bytes) used_bytes
        FROM dba_extents
        GROUP by file_id) E,
       (SELECT sum(bytes) free_bytes,
              file_id
        FROM dba_free_space
        GROUP BY file_id) f
WHERE  e.file_id (+) = df.file_id
AND    df.file_id = f.file_id (+)
ORDER BY df.tablespace_name,
        df.file_name;
```

Add Datafile NOXIOUS_TS

```
alter tablespace TS_NOXIOUS add datafile '/oradata/NOXIOUS/ts_noxious01.dbf' size 10m autoextend
on next 100m maxsize unlimited
```

CHECK SIZE DATABASE ORACLE

```
set lines 999
```

```
set pages 999
```

```
col HOST_NAME for a20
```

```
SELECT SYSDATE DATED, A.HOST_NAME, A.INSTANCE_NAME, A.VERSION, C.TOTAL_ALLOCATED_GB,
B.USED_GB, D.FREE_GB FROM V$INSTANCE A,
```

```
(SELECT SUM(BYTES)/1024/1024/1024 USED_GB FROM DBA_SEGMENTS) B,
```

```
(SELECT SUM(BYTES)/1024/1024/1024 TOTAL_ALLOCATED_GB FROM DBA_DATA_FILES) C,
```

```
(SELECT SUM(BYTES)/1024/1024/1024 FREE_GB FROM DBA_FREE_SPACE) D;
```

```
col "Database Size" format a20
```

```
col "Free space" format a20
```

```
col "Used space" format a20
```

```
select round(sum(used.bytes) / 1024 / 1024 / 1024 ) || ' GB' "Database Size"
```

```
, round(sum(used.bytes) / 1024 / 1024 / 1024 ) -
```

```
round(free.p / 1024 / 1024 / 1024) || ' GB' "Used space"
```

```
, round(free.p / 1024 / 1024 / 1024) || ' GB' "Free space"
```

```
from (select bytes
```

```
from v$datafile
```

```
union all
```

```
select bytes
```

```
from v$tempfile
```

```

union all
select bytes
from v$log) used
, (select sum(bytes) as p
from dba_free_space) free
group by free.p
/

```

COMMAND CHECK DIRECTORY LARGEST

```
du -sh * 2>/dev/null | sort -n -r | head -n 10
```

Query Check Usage Recovery_Dest

```

col name format a32
col size_mb format 999,999,999
col used_mb format 999,999,999
col pct_used format 999
select name, ceil( space_limit / 1024 / 1024) size_mb,
ceil( space_used / 1024 / 1024) used_mb, decode( nvl( space_used, 0),0, 0,
ceil ( ( space_used / space_limit) * 100) ) pct_used
from v$recovery_file_dest order by name desc;

```

Relocate Datafile To another location

Relocate datafile to another location (offline Mode)

[1] Check current location tablespace

```
SYS@Rptmvc> select name from v$datafile;
```

NAME

/oradata/RPSCMS/system01.dbf

/oradata/RPSCMS/sysaux01.dbf

/oradata/RPSCMS/undotbs01.dbf

/u01/app/oracle/product/19.0.0/dbhome_1/dbs/RPSCMS_TS.dbf

/oradata/RPSCMS/users01.dbf

[2] Set tablespace offline

```
SQL> ALTER TABLESPACE RPSCMS_TS OFFLINE NORMAL;
```

[3] move file physical to new location

```
mv '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/RPTMVC_TS.dbf'  
'/oradata/RPSCMS/RPSCMS_TS.dbf';
```

[4] Rename Tablespace

```
SQL> ALTER TABLESPACE RPSCMS_TS RENAME DATAFILE  
'/u01/app/oracle/product/19.0.0/dbhome_1/dbs/ RPSCMS_TS.dbf' TO '/oradata/RPSCMS/  
RPSCMS_TS.dbf';
```

[5] Set tablespace Online

```
SQL> ALTER TABLESPACE RPSCMS_TS ONLINE;
```

Relocate datafile to another location (online Mode)

```
=====
|| Relocate Datafiles to another Location ||
=====

[1] check current Tablespace location
    - select name from v$datafile;

SYS@RPTMVC> select name from v$datafile;

NAME
-----
/oradata/RPTMVC/system01.dbf
/oradata/RPTMVC/sysaux01.dbf
/oradata/RPTMVC/undotbs01.dbf
/u01/app/oracle/product/19.0.0/dbhome_1/dbs/RPTMVC_TS.dbf
/oradata/RPTMVC/users01.dbf

[2] shutdown instance
    - shu immediate

[3] move file physical from old location to new location
    - mv /u01/app/oracle/product/19.0.0/dbhome_1/dbs/RPTMVC_TS.dbf /oradata/RPTMVC/RPTMVC_TS.dbf

[4] check file exists on new location
    - ls -l /oradata/RPTMVC/RPTMVC_TS.dbf

[5] start instance mount
    - startup mount

[6] alter database rename file '/u01/app/oracle/product/19.0.0/dbhome_1/dbs/RPTMVC_TS.dbf' to '/oradata/RPTMVC/RPTMVC_TS.dbf';

[7] open database
    - alter database open

[8] check datafile location
    - select name from v$datafile;
```

KILL ACTIVE SESSION

Check SID and SERIAL Session

SQL> col SCHEMANAME for a20

col USERNAME for a20

```
select SQL_ID, SID, SERIAL#, STATUS, SCHEMANAME, USERNAME from v$session where SQL_ID='5svp8vrung9n4';
```

Kill Session

```
SQL> ALTER SYSTEM KILL SESSION 'sid,serial#' IMMEDIATE;
```

#888 = sid

#26215 = serial

```
SQL> ALTER SYSTEM KILL SESSION '16,2533' IMMEDIATE;
```

RESYNC / SYNC STANDBY DATAGUARD ORACLE DATABASE 19C

Archive PRIMARY

```

SYS@simiandb> archive log list
Database log mode           Archive Mode
Automatic archival          Enabled
Archive destination         /oraarc2/SIMIANDB
Oldest online log sequence  25668
Next log sequence to archive 25671
Current log sequence        25671
SYS@simiandb> █

```

Archive STANDBY

```

SYS@simiandb> archive log list
Database log mode           Archive Mode
Automatic archival          Enabled
Archive destination         /oraarc2/SIMIANDB
Oldest online log sequence  16940
Next log sequence to archive 0
Current log sequence        16943
SYS@simiandb> █

```

ON STANDBY

1. stop MRP
alter database recover managed standby database cancel;
2. masuk ke RMAN TARGET
\$ rman target /

RMAN> RECOVER STANDBY DATABASE FROM SERVICE **SIMIANDB_DC**;

- **SIMIANDB_DC = name tns primary**

```

[oracle@jtlmedalldb abas]$ rman target /

Recovery Manager: Release 19.0.0.0.0 - Production on Thu Oct 12 18:04:34 2023
Version 19.9.0.0.0

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connected to target database: SIMIANDB (DBID=2971151236, not open)

RMAN> RECOVER STANDBY DATABASE FROM SERVICE SIMIANDB_DC;

Starting recover at 12-OCT-23
Oracle instance started

Total System Global Area      1288489928 bytes

Fixed Size                      15988824 bytes
Variable Size                  2080374784 bytes
Database Buffers               10770972672 bytes
Redo Buffers                    17563648 bytes

contents of Memory Script:
{
  restore standby controlfile from service 'SIMIANDB_DC';
  alter database mount standby database;
}
executing Memory Script

Starting restore at 12-OCT-23
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=857 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: using network backup set from service SIMIANDB_DC
channel ORA_DISK_1: restoring control file
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
output file name=/oradata/SIMIANDB/control01.ctl
output file name=/oralog1/SIMIANDB/control02.ctl
output file name=/oralog2/SIMIANDB/control03.ctl
Finished restore at 12-OCT-23

released channel: ORA_DISK_1
Statement processed
For record type BACKUP PIECE RECID$ from 57568 to 75559 are re-used before resync

```

Jika terdapat error seperti berikut, cek incremetal pada primary, kemudian samakan status incarnation pada standby seperti pada primary.

```

starting media recovery
media recovery failed
RMAN-00571: =====
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of recover command at 10/12/2023 18:24:40
RMAN-03015: error occurred in stored script Memory Script
ORA-00283: recovery session canceled due to errors
RMAN-11003: failure during parse/execution of SQL statement: alter database recover
            if needed standby start
ORA-00283: recovery session canceled due to errors
ORA-19909: datafile 1 belongs to an orphan incarnation
ORA-01110: data file 1: '/oradata/SIMIANDB/system01.dbf'

RMAN> exit

Recovery Manager complete.

```

untuk kasus simiandb, karena status incarnation pada primary saat ini adalah 2, maka reset incarnation pada standby menjadi 2 juga.

```
[oracle@jtlmedalldb abas]$ rman target /
```

```
Recovery Manager: Release 19.0.0.0.0 - Production on Thu Oct 12 18:34:11 2023  
Version 19.9.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.
```

```
connected to target database: SIMIANDB (DBID=2971151236, not open)
```

```
RMAN> list incarnation;
```

```
using target database control file instead of recovery catalog
```

```
List of Database Incarnations
```

DB Key	Inc Key	DB Name	DB ID	STATUS	Reset SCN	Reset Time
1	1	SIMIANDB	2971151236	PARENT	1	17-APR-19
2	2	SIMIANDB	2971151236	PARENT	1920977	06-JAN-21
3	3	SIMIANDB	2971151236	ORPHAN	202629749	08-JUN-21
4	4	SIMIANDB	2971151236	ORPHAN	203618431	08-JUN-21
5	5	SIMIANDB	2971151236	CURRENT	1053124991	15-JUL-22

```
RMAN> reset database to incarnation 2;
```

```
database reset to incarnation 2
```

```
RMAN> exit
```

Nyalakan MRP

```
SQL> alter database recover managed standby database using current logfile disconnect from  
session;
```

Delete Archivelog

---- Cek size fra

```
COL NAME FORMAT A40
```

```
COL "Size MB" FORMAT 999,999,999
```

```
COL "Used MB" FORMAT 999,999,999.99
```

```
COL "% Free" FORMAT 999.99
```

```
COL "% Used" FORMAT 999.99
```

```
SELECT NAME,
```

```
    SPACE_LIMIT / 1024 / 1024 AS "Size MB",
```

```
    ROUND(SPACE_USED / 1024 / 1024, 2) AS "Used MB",
```

```
    ROUND((SPACE_LIMIT - SPACE_USED) / SPACE_LIMIT * 100, 2) AS "% Free",
```

```
    ROUND((SPACE_USED / SPACE_LIMIT) * 100, 2) AS "% Used"
```

```
FROM V$RECOVERY_FILE_DEST;
```

---- CEK SIZE DB FROM MB ----

```
SQL> SELECT sum(bytes)/1024/1024 "Total Size in MB" FROM dba_data_files;
```

---- CEK PARAMETER FAST RECOVERY AREA ----

```
SQL> show parameter recovery;
```

--- CEK ARCHIVE DESTINATION ----

```
SQL> archive log list;
```

---- HK ARCHIVELOG FRA SYSDATE BY OPTIONAL ----

rman target /

RMAN> crosscheck archivelog all;

RMAN> delete archivelog until time 'SYSDATE-3';

RMAN> delete noprompt archivelog until time 'SYSDATE-1';

RMAN> delete archivelog until sequence <Nomor Sequence>

RMAN> delete archivelog until time 'sysdate-1/24'

Case khusus delete archivelog tidak bisa seperti di RMTOOLDBDRC

1. Gunakan force

RMAN> delete force archivelog until sequence <NOMOR SEQUENCE>

RMAN> delete force archivelog until time 'sysdate-1'

RMAN> delete force archivelog until time 'sysdate-1/24'

```
RMAN> delete force archivelog until sequence 12409;

released channel: ORA_DISK_1
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=141 device type=DISK
List of Archived Log Copies for database with db_unique_name RMTDBDRC
=====

Key       Thrd Seq      S Low Time
-----
11506     1    12409    A 31-OCT-23
          Name: /oraarc1/rmt_1_12409_1118658903.arc

Do you really want to delete the above objects (enter YES or NO)? █
```

CHECK FREESPACE DISK GROUP ASM

```
SET ECHO    OFF
SET FEEDBACK 6
SET HEADING  ON
SET LINESIZE 180
SET PAGESIZE 50000
SET TERMOUT  ON
SET TIMING   OFF
SET TRIMOUT  ON
SET TRIMPOOL ON
SET VERIFY   OFF
CLEAR COLUMNS
CLEAR BREAKS
CLEAR COMPUTES
COLUMN group_name      FORMAT a25      HEAD 'Disk Group|Name'
COLUMN state           FORMAT a11      HEAD 'State'
COLUMN total_mb        FORMAT 999,999,999  HEAD 'Total Size (GB)'
COLUMN free_mb         FORMAT 999,999,999  HEAD 'Free Size (GB)'
COLUMN pct_free        FORMAT 999.99      HEAD '% Free'
BREAK ON report ON disk_group_name SKIP 1
COMPUTE sum LABEL "Grand Total: " OF total_mb used_mb ON report
SELECT
    name                group_name
  , state                state
  , total_mb/1024        total_mb
  , free_mb/1024         free_mb
  , ROUND((((free_mb / total_mb))*100, 2)  pct_free
```

```
FROM
    v$asm_diskgroup
WHERE
    total_mb != 0
ORDER BY
    Name
/
```

check usage asm disk group

```
SET ECHO    OFF
SET FEEDBACK 6
SET HEADING  ON
SET LINESIZE 180
SET PAGESIZE 50000
SET TERMOUT  ON
SET TIMING   OFF
SET TRIMOUT  ON
SET TRIMSPOOL ON
SET VERIFY   OFF

CLEAR COLUMNS
CLEAR BREAKS
CLEAR COMPUTES

COLUMN group_name      FORMAT a25      HEAD 'Disk Group|Name'
COLUMN state           FORMAT a11      HEAD 'State'
COLUMN total_mb        FORMAT 999,999,999  HEAD 'Total Size (GB)'
COLUMN used_mb         FORMAT 999,999,999  HEAD 'Used Size (GB)'
COLUMN pct_used        FORMAT 999.99      HEAD '% Used'
```



```

BREAK ON report ON disk_group_name SKIP 1

COMPUTE sum LABEL "Grand Total: " OF total_mb used_mb ON report

SELECT
    name                group_name
, state                state
, total_mb/1024        total_mb
, (total_mb - free_mb)/1024    used_mb
, ROUND((1- (free_mb / total_mb))*100, 2) pct_used
FROM
    v$asm_diskgroup

WHERE
    total_mb != 0

ORDER BY
    Name
/

```

Purge Log

masuk ke file .log (biasanya di trace) jalankan

```
> nama.log
```

```

/dev/mapper/data-d01-1000-050-100-050 /d01
[oracle@MVRKBETADB01 trace]$ hostname
MVRKBETADB01
[oracle@MVRKBETADB01 trace]$ > sbtio.log

```

ADD DATAFILE TABLESPACE TEMP

1. check dahulu lokasi datafile temp nya dengan query dibawah

```
set pages 999

set lines 400

col FILE_NAME format a75

select d.TABLESPACE_NAME, d.FILE_NAME, d.BYTES/1024/1024 SIZE_MB,
d.AUTOEXTENSIBLE, d.MAXBYTES/1024/1024 MAXSIZE_MB,
d.INCREMENT_BY*(v.BLOCK_SIZE/1024)/1024 INCREMENT_BY_MB
from dba_temp_files d,
v$tempfile v
where d.FILE_ID = v.FILE#
order by d.TABLESPACE_NAME, d.FILE_NAME;
```

```
TABLESPACE_NAME FILE_NAME SIZE_MB AUT MAXSIZE_MB INCREMENT_BY_MB
-----
TEMP +DATAAC1/VERS/TEMPFILE/temp.451.891367325 32767 YES 32767 1024
TEMP +DATAAC1/VERS/TEMPFILE/temp.452.891367321 32767 YES 32767 1024
TEMP +DATAAC1/VERS/TEMPFILE/temp.454.891367305 32767 YES 32767 1024
```

2. tambah datafile dengan command dibawah

```
ALTER TABLESPACE TEMP ADD TEMPFILE '+DATAAC1' SIZE 10G AUTOEXTEND ON NEXT
1G MAXSIZE 32767M;
```

notes : meskipun file lokasi bearada di +DATAAC1/VERS/TEMPFILE namun yang di mention pada script alter cukup mountingan saja yaitu +DATAAC1. seperti command diatas

HOW TO LOGIN ASM

1. untuk server yang menggunakan asm maka mengecek file yang berada di ASM directory yaitu dengan menggunakan command

~\$ asmcmd

~\$ ls -lrt

```
[oracle@NEWCOREDBUAT01 ~]$ asmcmd
ASMCMD> ls -lrt
State      Type      Rebal     Name
MOUNTED    EXTERN    N         RECO2/
MOUNTED    EXTERN    N         RECO/
MOUNTED    EXTERN    N         DATA2/
MOUNTED    EXTERN    N         DATA/
ASMCMD>
```

notes : jangan lupa act dahulu seperti biasa.

CREATE RESTORE POINT

-- CEK APAKAH FLASHBACK SUDAH AKTIF

```
SQL> SELECT flashback_on FROM v$database;
```

-- Jika hasilnya = 'NO', maka aktifkan flashback (di langkah [3] setelah mount)

CEK DAN SET FRA (Flash Recovery Area) JIKA BELUM ADA

-- Cek dulu

```
SQL> SHOW PARAMETER db_recovery_file_dest;
```

```
SQL> SHOW PARAMETER db_recovery_file_dest_size;
```

-- Jika BELUM diset, maka set seperti ini:

```
SQL> ALTER SYSTEM SET db_recovery_file_dest = '/backup/fra' SCOPE=BOTH;
```

```
SQL> ALTER SYSTEM SET db_recovery_file_dest_size = 200G SCOPE=BOTH;
```

AKTIFKAN FLASHBACK DATABASE JIKA BELUM

-- Harus dalam mode MOUNT

```
SQL> SHUTDOWN IMMEDIATE;
```

```
SQL> STARTUP MOUNT;
```

```
SQL> ALTER DATABASE FLASHBACK ON;
```

```
SQL> ALTER DATABASE OPEN;
```

-- BUAT GUARANTEED RESTORE POINT

```
SQL> CREATE RESTORE POINT avq_170824 GUARANTEE FLASHBACK DATABASE;
```

```
-- CEK RESTORE POINT YANG ADA
```

```
SET LINES 200
```

```
COL name FOR A30
```

```
COL time FOR A35
```

```
SELECT con_id, name, time, guarantee_flashback_database
```

```
FROM v$restore_point
```

```
ORDER BY 1,2;
```

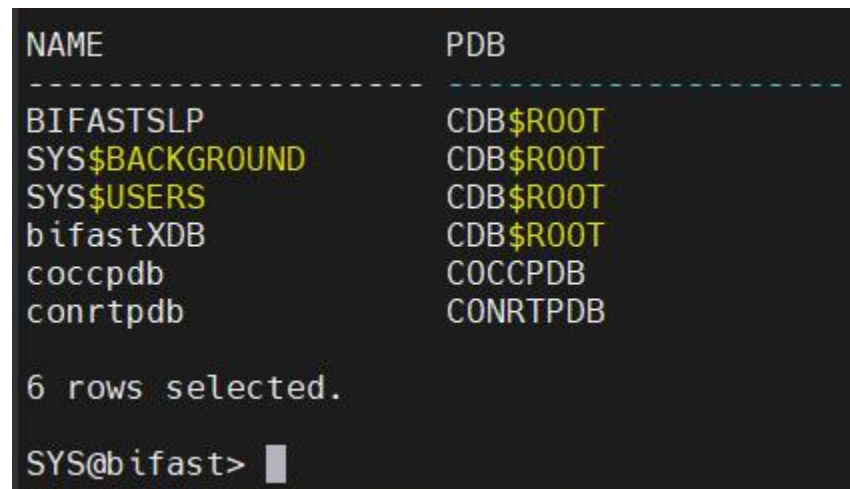
CEK TABLESPACE CONTAINER

cek container pada tablespaces

col name for a20

col pdb for a20

select name , pdb from v\$tablespaces order by name;



The screenshot shows a SQL query result with two columns: NAME and PDB. The results are as follows:

NAME	PDB
BIFASTSLP	CDB\$ROOT
SYS\$BACKGROUND	CDB\$ROOT
SYS\$USERS	CDB\$ROOT
bifastXDB	CDB\$ROOT
coccpdb	COCCPDB
conrtpdb	CONRTPDB

6 rows selected.

SYS@bifast>

alter session set container=CONRTPDB;

alter session set container=CONRTPDB;

jalankan alter di atas di masing masing db CCRTDBDC01 -> CONRTPDB

CCNRTUIDBDC -> CONRTPDB

setelah jalankan alter baru melakukan cek tablespace menggunakan query di bawah

SET LINESIZE 200;

SET PAGESIZE 999;

COLUMN name FOR a10;

```
COLUMN tablespace_name FOR a15;  
COLUMN "MAXSIZE (MB)" FORMAT 9,999,990.00;  
COLUMN "ALLOC (MB)" FORMAT 9,999,990.00;  
COLUMN "USED (MB)" FORMAT 9,999,990.00;  
COLUMN "PERC_USED" FORMAT 99.00;
```

```
SELECT a.con_id,  
       c.name,  
       b.tablespace_name,  
       a.bytes_alloc / (1024 * 1024) AS "MAXSIZE (MB)",  
       NVL(a.physical_bytes, 0) / (1024 * 1024) AS "ALLOC (MB)",  
       NVL(b.tot_used, 0) / (1024 * 1024) AS "USED (MB)",  
       (NVL(b.tot_used, 0) / a.bytes_alloc) * 100 AS "PERC_USED"  
FROM (SELECT con_id,  
            tablespace_name,  
            SUM(bytes) AS physical_bytes,  
            SUM(DECODE(autoextensible, 'NO', bytes, 'YES', maxbytes)) AS bytes_alloc  
      FROM cdb_data_files  
      GROUP BY con_id, tablespace_name) a  
LEFT JOIN (SELECT con_id,  
                tablespace_name,  
                SUM(bytes) AS tot_used  
          FROM cdb_segments  
          GROUP BY con_id, tablespace_name) b  
ON a.con_id = b.con_id AND a.tablespace_name = b.tablespace_name  
JOIN v$containers c  
ON a.con_id = c.con_id  
ORDER BY a.con_id, b.tablespace_name;
```

Lokasi path / direktori / folder alert log

login ke sqlplus : lalu jalankan query dibawah

```
select value from v$diag_info;
```

CEK ALERT LOG

locate alert_\${ORACLE_SID}.log

CREATE USER IN DATABASE

CREATE USER nama_user IDENTIFIED BY password;

CREATE USER noxious IDENTIFIED BY 123;

CEK USER YANG ADA

SELECT * FROM all_users;

CEK USER YANG SEDANG AKTIF

SELECT DISTINCT USERNAME FROM V\$SESSION WHERE USERNAME IS NOT NULL;

MEMBERI AKSES MEMBACA SEMUA TABLE

GRANT READ ANY TABLE TO noxious;

CEK SCHEMA USER ORACLE

select username as schema_name from sys.all_users order by username;

CEK USER DILUAR BAWAAN ORACLE

select username as schema_name from sys.all_users where ORACLE_MAINTAINED = 'N';

CEK TABLE

```
select table_name from all_tables;
```

```
SELECT table_name FROM all_tables WHERE owner ='nama_user';
```

```
SELECT table_name FROM all_tables WHERE owner ='DBADMIN';
```

```
SELECT 'SELECT COUNT(*) FROM ' || owner || ' ' || table_name || ';' AS sql_query
```

```
FROM all_tables
```

```
WHERE owner = 'DBADMIN';
```

```
SELECT OWNER, table_name FROM all_tables WHERE table_name = '%TXT%' OR table_name  
='%txt%';
```

```
SELECT OWNER, table_name FROM all_tables WHERE table_name = 'TXTYPE';
```

CEK OWNER TABLE

```
Select owner from dba_tables where table_name = 'TXTYPE';
```

```
Select owner from dba_tables where table_name = 'MGMT_DB_FILE_GTT';
```

CEK BERAPA BANYAK ISI TABLE

```
select count (1) FROM BTN2.TXTYPE
```

MEMBERI AKSES ALL

```
GRANT ALL PRIVILEGES TO nama_user;
```

```
GRANT ALL PRIVILEGES TO NOXIOUS;
```

MENGHAPUS USER

```
DROP USER nama_user;
```

MEMBERI AKSES UNTUK CONNECT USER

```
GRANT CONNECT TO nama_user;
```

```
GRANT CONNECT TO rex;
```

MEMBERI AKSES UNTUK SELECT SEMUA TABLE

```
GRANT SELECT ANY TABLE TO nama_user;
```

```
GRANT SELECT ANY TABLE TO rex;
```

CEK ISI TABLE

```
select * from namaschema.namatable;
```

```
contoh : select * from BTN2.TXTYPE;
```

MASUK KE DALAM PDBS

```
ALTER SESSION SET CONTAINER = PKONVRT;
```

```
ALTER SESSION SET CONTAINER = PKONVNRT;
```

```
ALTER SESSION SET CONTAINER = PSHADNRT;
```

```
ALTER SESSION SET CONTAINER = PSHADRT;
```

CEK PDBS

```
show pdbs;
```

CEK USER EXPIRED

(EXPIRED)

```
SQL> select username, account_status, expiry_date from dba_users;
```

(DEFAULT TABLESPACE)

```
SQL> select username, account_status, default_tablespace from dba_users;
```

CEK PRIVILEGES GRANT

```
select * from dba_sys_privs;
```

```
select * from dba_sys_privs order by grantee;
```

```
select * from dba_sys_privs where grantee='user_secure' order by grantee;
```

CHANGE SIZE SGA & PGA

```
ALTER SYSTEM SET SGA_TARGET = 1504M;
```

```
ALTER SYSTEM SET SGA_MAX_SIZE = 1000M scope=spfile;
```

```
alter system set sga_max_size = 12G scope=spfile;
```

CHANGE DB UNIQ NAME

```
ALTER SYSTEM SET DB_UNIQUE_NAME='sbyorcl' SCOPE=SPFILE;
```

```
srvctl start instance -d DBWMS -i dbwms1 -o mount
```

```
srvctl start instance -d DBATMINS -i DBATMINS1 -o mount
```

CEK USER KAPAN DIBUAT

```
SELECT USERNAME, CREATED
```

```
FROM DBA_USERS AS OF TIMESTAMP (SYSTIMESTAMP - INTERVAL '1' DAY)
```

```
WHERE USERNAME = 'BIMO';
```

COMPARE DATA DC DAN DRC

```
select count(table_name) from all_tables;

select sum(bytes)/1024/1024/1024 from dba_segments;

select sum(bytes)/1024/1024/1024 from dba_data_files;
```

Cek path Datafile, logfile, tempfile, archive

```
SELECT NAME FROM V$DATAFILE;

SELECT MEMBER FROM V$LOGFILE;

SELECT FILE_NAME FROM DBA_TEMP_FILES;

SHOW PARAMETER LOG_ARCHIVE_DEST;

SELECT DEST_NAME, DESTINATION FROM V$ARCHIVE_DEST;
```

ALERT DATAGUARD

```
set pagesize 9999

set linesize 200

select message from v$dataguard_status;


set pagesize 9999

set linsize 200

alter session set nls_date_format='dd-mm-yyyy hh24:mi:ss';

select message from v$dataguard_status;
```

APPLIED SEQUENCE

```
select * from (select thread#, max(sequence#), APPLIED from v$archived_log where
first_time>=sysdate-5 group by thread#,

first_time, sequence#, APPLIED order by first_time desc) where rownum <=10 and APPLIED='YES';
```

```
select * from (select thread#, max(sequence#), APPLIED from v$archived_log where  
first_time>=sysdate-5 group by thread#,  
first_time, sequence#, APPLIED order by first_time desc) where rownum <=50;
```

CEK STATUS APPLIED

```
SQL> select process, status, thread#, sequence# from v$managed_standby;
```

CEK TEMPFILE

```
SQL> select name, FILE# from v$tempfile;
```

MEMBUAT TEMP FILE

```
SQL> show pdbs;
```

```
SQL> alter session set container=PKONVNRT;
```

```
SQL> select name, FILE# from v$tempfile;
```

```
SQL> alter tablespace TEMP add tempfile size 8192M autoextend on next 100M;
```

```
SQL> select name, FILE# from v$tempfile;
```

MEMBUAT PFILE

```
SQL> create pfile='<pfile location>' from spfile;
```

```
SQL> create pfile='/home/oracle/pfile_orcl.ora' from spfile;
```

MEMBUAT SPFILE DARI PFILE

```
SQL> create spfile from pfile = '<pfile location>;
```

STARTUP SPFILE & PFILE

```
SQL> startup pfile='<pfile location><nama pfile>;
```

```
SQL> startup spfile='<spfile location>;
```

CEK SPFILE & PFILE

```
SQL> show parameter spfile;
```

```
SQL> show parameter pfile;
```

CEK STATUS BACKUP RMAN

```
SQL> set linesize 500 pagesize 2000
```

```
col Hours format 9999.99
```

```
col STATUS format a10
```

```
select SESSION_KEY, INPUT_TYPE, STATUS,
```

```
to_char(START_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_start_time,
```

```
to_char(END_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_end_time, elapsed_seconds/3600  
Hours from V$RMAN_BACKUP_JOB_DETAILS order by session_key;
```

```
set linesize 500 pagesize 2000
```

```
col Hours format 9999.99
```

```
col STATUS format a10
```

```
select SESSION_KEY, INPUT_TYPE, STATUS,
```

```
to_char(START_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_start_time,
```

```
to_char(END_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_end_time, elapsed_seconds/3600  
Hours from V$RMAN_BACKUP_JOB_DETAILS
```

```
where STATUS='COMPLETED'
```

```
order by session_key desc;
```

CEK STATUS BACKUP RMAN & SIZE BACKUP

```
set linesize 500
```

```
col BACKUP_SIZE for a20
```

```

SELECT
INPUT_TYPE "BACKUP_TYPE",
--NVL(INPUT_BYTES/(1024*1024),0)"INPUT_BYTES(MB)",
--NVL(OUTPUT_BYTES/(1024*1024),0) "OUTPUT_BYTES(MB)",
STATUS,
TO_CHAR(START_TIME,'MM/DD/YYYY:hh24:mi:ss') as START_TIME,
TO_CHAR(END_TIME,'MM/DD/YYYY:hh24:mi:ss') as END_TIME,
TRUNC((ELAPSED_SECONDS/60),2) "ELAPSED_TIME(Min)",
--ROUND(COMPRESSION_RATIO,3)"COMPRESSION_RATIO",
--ROUND(INPUT_BYTES_PER_SEC/(1024*1024),2) "INPUT_BYTES_PER_SEC(MB)",
--ROUND(OUTPUT_BYTES_PER_SEC/(1024*1024),2) "OUTPUT_BYTES_PER_SEC(MB)",
--INPUT_BYTES_DISPLAY "INPUT_BYTES_DISPLAY",
OUTPUT_BYTES_DISPLAY "BACKUP_SIZE",
OUTPUT_DEVICE_TYPE "OUTPUT_DEVICE"
--INPUT_BYTES_PER_SEC_DISPLAY "INPUT_BYTES_PER_SEC_DIS",
--OUTPUT_BYTES_PER_SEC_DISPLAY "OUTPUT_BYTES_PER_SEC_DIS"
FROM V$RMAN_BACKUP_JOB_DETAILS
where start_time > SYSDATE -10
and INPUT_TYPE != 'ARCHIVELOG'
ORDER BY END_TIME DESC
/

```

House Keeping File

(HARUS BERADA DI DALAM PATH YANG MAU DI HK)

(UNTUK LIST)

```
/usr/bin/find ./ -iname "*.bkp*" -mtime +3 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname "*.trc*" -mtime +3 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname "*.tr*" -mtime +3 -exec ls -lrt {} \;
```

(UNTUK HOUSE KEEPING)

```
/usr/bin/find ./ -iname "*.bkp*" -mtime +3 -exec rm -rf {} \;
```

```
/usr/bin/find ./ -iname "*.trc*" -mtime +3 -exec rm -rf {} \;
```

```
/usr/bin/find ./ -iname "*.tr*" -mtime +3 -exec rm -rf {} \;
```

---- HOUSEKEEPING EXADATA BTN ----

--- .xml ---

--- log database format "tr"

```
sql> select value from V$DIAG_INFO WHERE name='Diag Trace';
```

--- log database format "aud"

```
sql> show parameter audit
```

(PASTIKAN SUDAH ADA DI PATH TERSEBUT!!)

```
/usr/bin/find ./ -iname "*.trc" -mtime +7 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname "*.trc" -mtime +7 -exec rm -rf {} \;
```

```
/usr/bin/find ./ -iname ".aud" -mtime +14 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname ".aud" -mtime +14 -exec rm -rf {} \;
```

```
/usr/bin/find ./ -iname "*.trc" -mtime +7 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname "*.trc" -mtime +7 -exec rm -rf {} \;
```

```
/usr/bin/find ./ -iname "*aud" -mtime +1 -exec ls -lrt {} \;
```

```
/usr/bin/find ./ -iname "*aud" -mtime +1 -exec rm -rf {} \;
```


CEK DB LINK

SET LINESIZE 150

COLUMN owner FORMAT A30

COLUMN db_link FORMAT A30

COLUMN username FORMAT A30

COLUMN host FORMAT A30

SELECT owner, db_link, username, host FROM dba_db_links;

CEK DB LINK 2

SELECT DB_LINK, USERNAME, HOST FROM ALL_DB_LINKS;

CEK STATUS VALID DB LINK

SELECT DB_LINK, OWNER, HOST, VALID FROM DBA_DB_LINKS;

MENGHAPUS DB LINK

DROP DATABASE LINK dblink;

MENGUBAH OWNER ORACLE

(MENGUNAKAN ROOT)

chown (new owner)(filename)

chown oracle:oinstall (nama file)

CEK LOCKING

select c.owner, c.object_name, c.object_type, b.sid, b.serial#, b.status, b.osuser, b.machine from
v\$locked_object a,

v\$session b, dba_objects c where b.sid = a.session_id and a.object_id = c.object_id;

CEK STATUS BACKUP RMAN

```
set linesize 500 pagesize 2000

col Hours format 9999.99

col STATUS format a10

select SESSION_KEY, INPUT_TYPE, STATUS,
to_char(START_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_start_time,
to_char(END_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_end_time, elapsed_seconds/3600
Hours from V$RMAN_BACKUP_JOB_DETAILS

where STATUS='COMPLETED'

order by session_key desc;
```

```
set linesize 500 pagesize 2000

col Hours format 9999.99

col STATUS format a10

select SESSION_KEY, INPUT_TYPE, STATUS,
to_char(START_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_start_time,
to_char(END_TIME,'mm-dd-yyyy hh24:mi:ss') as RMAN_Bkup_end_time, elapsed_seconds/3600
Hours from V$RMAN_BACKUP_JOB_DETAILS

order by session_key desc;
```

BACKUP RMAN

--- Melakukan backup database pada server prod jalankan script sh backup yang tersedia

```
#!/bin/bash

export DATETIME=$(date +%d%m%y_%H%M%S)

export ORACLE_SID=orcl

#export ORACLE_BASE=/u01/app/oracle

export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1

export ORACLE_HOSTNAME=exaprimary

export PATH=$PATH:$ORACLE_HOME/bin


${ORACLE_HOME}/bin/rman target=/
log=/home/oracle/backup/log/alertbackupORCL_${BCKDate}.log << EOF

run

{

crosscheck backup;

crosscheck archivelog all;

delete noprompt expired archivelog all;

delete noprompt expired backup;

backup as compressed backupset incremental level 0 database format
'/home/oracle/backup/bckup/Backup_May/data_level0_%d_%s_%p_%c_%T.bkp';

backup as compressed backupset incremental level 0 archivelog all format
'/home/oracle/backup/bckup/Backup_May/archive_level0_%d_%s_%p_%c_%T.bak';

backup current controlfile for standby format
'/home/oracle/backup/bckup/Backup_May/exa_controlfile_level0_%d_%s_%p_%c_%T.bkp';

backup current controlfile format
'/home/oracle/backup/bckup/Backup_May/current_controlfile_level0_%d_%s_%p_%c_%T.bkp';

}

EXIT;

EOF
```

--- Membuat pfile dari server prod

create pfile='/home/oracle/prdkrt13032024.ora' from spfile; (hanya contoh penamaan nama dan lokasi)

create pfile='/home/oracle/orclprim.ora' from spfile;

--- SCP FILE HASIL BACKUP DARI PROD KE DEV

scp prdkrt13032024.ora oracle@172.15.3.137:/temporary/BACKUPPRDKRT

--- Melakukan drop database pada server dev

sql> shutdown immediate

sql> startup mount exclusive restrict

sql> drop database

RESTORE BACKUP RMAN

(NOTES : PASTIKAN ARCHIVELOG & DATAFILE CLEAR)

--- startup menggunakan pfile yang telah dibackup dari prod

```
sql> startup nomount pfile='/temporary/BACKUPPRDKRT/prdkrt13032024.ora';
```

--- restore controlfile

```
rman target /
```

```
rman> restore controlfile from
```

```
'/temporary/dbrtgs/rman_20240517/DBRTGS_CTL_20240517_og2r1v7o_1_1.rman'; (sesuaikan path dan nama control file)
```

```
rman> alter database mount;
```

--- restore database

```
rman> report schema;
```

```
rman> crosscheck backup;
```

```
rman> catalog start with '/home/oracle/backup/BCKUP/dbwms/Backup_Juni';
```

```
rman>
```

```
run {
```

```
SET NEWNAME FOR DATABASE TO 'C:\Users\noxious\backup\DATA_FILE\%b';
```

```
SET NEWNAME FOR tempfile 1 TO 'C:\Users\noxious\backup\DATA_FILE\%b';
```

```
restore database;
```

```
switch datafile all;
```

```
switch tempfile all;
```

```
}
```

--- recover database

```
rman target /
```

```
run {  
recover database;  
}
```

```
rman> alter database open RESETLOGS;
```

CEK DATABASE SETELAH RESTORE

```
sql> SELECT NAME, OPEN_MODE, DATABASE_ROLE FROM V$DATABASE;  
sql> create spfile from pfile;
```

BACKUP EXPDP

1. Pastikan Oracle Environment Siap

- Tersedia user dengan role DBA atau punya privilege DATAPUMP_EXP_FULL_DATABASE dan DATAPUMP_IMP_FULL_DATABASE.
- Pastikan Oracle listener dan database sudah running.
- Pastikan Oracle Data Pump Utility (expdp dan impdp) tersedia di PATH.

2. Cek/Tentukan Direktori untuk Backup

--Buat directory object di Oracle:

```
CREATE OR REPLACE DIRECTORY dp_dir AS '/u01/backup/datapump';
```

--Beri privilege ke user:

```
GRANT READ, WRITE ON DIRECTORY dp_dir TO your_user;
```

Notes: Gantilah /u01/backup/datapump sesuai lokasi penyimpanan yang Anda inginkan dan pastikan OS user Oracle punya akses ke folder tersebut.

LANGKAH BACKUP (EXPDP)

Backup Full Database

```
expdp your_user/your_password FULL=Y DIRECTORY=dp_dir DUMPFILE=full_backup_%U.dmp  
LOGFILE=full_backup.log FILESIZE=2G
```

Backup Per Schema

```
expdp your_user/your_password SCHEMAS=SCOTT DIRECTORY=dp_dir  
DUMPFILE=scott_backup.dmp LOGFILE=scott_backup.log
```

Backup Per Table

```
expdp your_user/your_password TABLES=SCOTT.EMP,SCOTT.DEPT DIRECTORY=dp_dir  
DUMPFILE=tables_backup.dmp LOGFILE=tables_backup.log
```

LANGKAH RESTORE (IMPDP)

Restore Full Database

```
impdp your_user/your_password FULL=Y DIRECTORY=dp_dir DUMPFILE=full_backup_%U.dmp  
LOGFILE=imp_full.log
```

Restore Schema

```
impdp your_user/your_password SCHEMAS=SCOTT DIRECTORY=dp_dir  
DUMPFILE=scott_backup.dmp LOGFILE=imp_scott.log
```

Restore Table

```
impdp your_user/your_password TABLES=SCOTT.EMP,SCOTT.DEPT DIRECTORY=dp_dir  
DUMPFILE=tables_backup.dmp LOGFILE=imp_tables.log
```

Remap Schema atau Tablespace (Opsional)

```
impdp your_user/your_password SCHEMAS=SCOTT REMAP_SCHEMA=SCOTT:SCOTT_NEW  
DIRECTORY=dp_dir DUMPFILE=scott_backup.dmp LOGFILE=imp_remap.log
```

```
impdp your_user/your_password FULL=Y REMAP_TABLESPACE=USERS:NEW_USERS  
DIRECTORY=dp_dir DUMPFILE=full_backup.dmp LOGFILE=imp_remap_ts.log
```


KONFIGURASI ACTIVE DATAGUARD (ADG)

SET LISTENER DAN TNSNAME PADA PRIMARY DAN STANDBY

TNSNAMES PRIMARY

```
[oracle@exaprimary admin]$ cat tnsnames.ora
```

```
# tnsnames.ora Network Configuration File:  
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora  
  
# Generated by Oracle configuration tools.  
  
## EDIT BY NOXIOUS FOR ADG
```

```
orcl_primary =  
    (DESCRIPTION =  
        (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.10)(PORT = 1521))  
        (CONNECT_DATA =  
            (SERVICE_NAME = orcl_primary)  
        )  
    )
```

```
orcl_standby =  
    (DESCRIPTION =  
        (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.20)(PORT = 1521))  
        (CONNECT_DATA =  
            (SERVICE_NAME = orcl_standby)  
        )  
    )
```

LISTENER PRIMARY

[oracle@exaprimary admin]\$ cat listener.ora

EDIT BY NOXIOUS FOR ADG

LISTENER_ORCL =

(DESCRIPTION_LIST =

(DESCRIPTION =

(ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.10)(PORT = 1521))

)

)

SID_LIST_LISTENER_ORCL =

(SID_LIST =

(SID_DESC =

(GLOBAL_DBNAME = orcl_primary)

(ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)

(SID_NAME = orcl)

)

)

[oracle@exaprimary admin]\$

TNSNAMES STANDBY

```
[oracle@exastandby dbhome_1]$ cd network/admin/
```

```
[oracle@exastandby admin]$ cat tnsnames.ora
```

```
# tnsnames.ora Network Configuration File:
```

```
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora
```

```
# Generated by Oracle configuration tools.
```

```
## EDIT BY NOXIOUS FOR ADG
```

```
LISTENER_ORCL =
```

```
    (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.20)(PORT = 1521))
```

```
ORCL_PRIMARY =
```

```
    (DESCRIPTION =
```

```
        (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.10)(PORT = 1521))
```

```
        (CONNECT_DATA =
```

```
            (SERVICE_NAME = orcl_primary)
```

```
        )
```

```
    )
```

```
ORCL_STANDBY =
```

```
    (DESCRIPTION =
```

```
        (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.20)(PORT = 1521))
```

```
        (CONNECT_DATA =
```

```
            (SERVER = DEDICATED)
```

```
            (SERVICE_NAME = orcl_standby)
```

```
        )
```

```
    )
```

LISTENER STANDBY

[oracle@exastandby admin]\$ cat listener.ora

EDIT BY NOXIOUS FOR ADG

LISTENER_ORCL =

(DESCRIPTION_LIST =

(DESCRIPTION =

(ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.20)(PORT = 1521))

)

)

SID_LIST_LISTENER_ORCL =

(SID_LIST =

(SID_DESC =

(GLOBAL_DBNAME = orcl_standby)

(ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)

(SID_NAME = orcl)

)

)

SET PARAMETER DB_UNIQUE_NAME DI KEDUA SERVER

```
SQL> ALTER SYSTEM SET db_unique_name='orcl1_primary' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET db_unique_name='orcl1_standby' SCOPE=SPFILE;
```

ENABLE ARCHIVE MODE DI KEDUA SERVER

```
SQL> ARCHIVE LOG LIST;
```

```
SQL> SHUTDOWN IMMEDIATE;
```

```
SQL> STARTUP MOUNT;
```

```
SQL> ALTER DATABASE ARCHIVELOG;
```

```
SQL> ALTER DATABASE OPEN;
```

KONFIGURASI PARAMETER DI PRIMARY

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_CONFIG='DG_CONFIG=(orcl_primary,orcl_standby)' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/archive VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=orcl_standby' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_FORMAT='%t_%s_%r.arc' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_MAX_PROCESSES=4 SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET DB_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET STANDBY_FILE_MANAGEMENT=AUTO SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET FAL_SERVER=orcl_standby SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET FAL_CLIENT=orcl_primary SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE SCOPE=SPFILE;
```

KONFIGURASI PARAMETER DI STANDBY

```
SQL> ALTER SYSTEM SET DB_UNIQUE_NAME='orcl_standby' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_CONFIG='DG_CONFIG=(orcl_primary,orcl_standby)'  
SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/archive  
VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=orcl_primary' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_2='SERVICE=orcl_primary ASYNC  
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=orcl_primary'  
SCOPE=BOTH;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_FORMAT='%t_%s_%r.arc' SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_MAX_PROCESSES=4 SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET  
DB_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'  
SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET  
LOG_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'  
SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET STANDBY_FILE_MANAGEMENT=AUTO SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET FAL_SERVER=orcl_primary SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET FAL_CLIENT=orcl_standby SCOPE=SPFILE;
```

```
SQL> ALTER SYSTEM SET REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE SCOPE=SPFILE;
```

*****NOTES KALAU EROR*****

SQL> show parameter local_listener;

NAME	TYPE	VALUE
------	------	-------

local_listener	string	
----------------	--------	--

SQL> ALTER SYSTEM SET
LOCAL_LISTENER='(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.56.10)(PORT=1522))'
SCOPE=BOTH;

-- di kedua server password sys harus sama (OPTIONAL)

ALTER USER sys IDENTIFIED BY oracle;
ALTER USER sys IDENTIFIED BY We1come#;

-- Pembuatan Password File di Primary (OPTIONAL)

orapwd file=\$ORACLE_HOME/dbs/orapwdbwms password=oracle force=y
orapwd file=\$ORACLE_HOME/dbs/orapwdbwms password=We1come# force=y

-- Cek Password File di Standby

ls -l \$ORACLE_HOME/dbs/orapw*

-- Salin Password File dari Primary ke Standby

scp orapwdbwms oracle@192.168.56.20:\$ORACLE_HOME/dbs/

-- Tes akses ke sqlplus lewat standby

sqlplus sys/We1come#@dbwms_standby as sysdba

-- bersihkan file di standby sebelum melakukan duplicate

```
rm -rf /u01/app/oracle/oradata/ORCL/*
```

```
rm -rf /u02/oradata/orcl_standby/*
```

```
rm -rf /u01/app/oracle/fast_recovery_area/ORCL/*
```

-- edit pfile initorcl.ora di standby Tambahkan atau pastikan parameter ini ada

```
db_name=ORCL
```

```
db_unique_name=orcl_standby
```

```
db_file_name_convert='/u01/app/oracle/oradata/ORCL','/u02/oradata/orcl_standby'
```

```
log_file_name_convert='/u01/app/oracle/oradata/ORCL','/u02/oradata/orcl_standby'
```

-- setelah itu up database ke nomount di standby

```
startup nomount;
```

```
startup nomount pfile='/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initdbwms.ora';
```

-- masuk ke rman di standby untuk memulai duplicate database

```
rman target sys/We1come#@dbwms_primary auxiliary sys/We1come#@dbwms_standby
```

```
rman target sys/oracle@orcl_primary auxiliary sys/oracle@orcl_standby
```

-- memulai duplicate database di standby

```
RMAN> DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE DATABASE
```

```
DORECOVER
```

```
SPFILE
```

```
SET DB_UNIQUE_NAME='orcl_standby'
```

```
SET CONTROL_FILES='/u02/oradata/orcl_standby/control01.ctl','/u02/oradata/orcl_standby/control02.ctl'
```

```
SET DB_FILE_NAME_CONVERT='/u01/app/oracle/oradata/ORCL','/u02/oradata/orcl_standby'
```

```
SET LOG_FILE_NAME_CONVERT='/u01/app/oracle/oradata/ORCL','/u02/oradata/orcl_standby'
```

```
SET LOG_ARCHIVE_DEST_1='LOCATION=/u02/oradata/orcl_standby/archivelog'
```

```
NOFILENAMECHECK;
```


-- DUPLICATE DATABASE DI STANDBY

RMAN> DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE DATABASE

DORECOVER

SPFILE

SET DB_UNIQUE_NAME='dbwms_standby'

SET

CONTROL_FILES='/u02/oradata/dbwms_standby/control01.ctl','/u02/oradata/dbwms_standby/control02.ctl'

SET DB_FILE_NAME_CONVERT='/u02/oradata/dbwms_primary','/u02/oradata/dbwms_standby'

SET LOG_FILE_NAME_CONVERT='/u02/oradata/dbwms_primary','/u02/oradata/dbwms_standby'

SET LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/archive/dbwms'

NOFILENAMECHECK;

-- setelah duplicate selesai naikan database ke read only di standby

SQL> ALTER DATABASE OPEN READ ONLY;

--Agar lebih mudah dalam mengelola Data Guard, gunakan Data Guard Broker Aktifkan Data Guard Broker di Primary dan Standby.

SQL> ALTER SYSTEM SET DG_BROKER_START=TRUE;

-- lalu menyalakan mrp pada standby jika belum nyala

SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;

atau bisa gunakan ini

SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT USING CURRENT
LOGFILE;

-- cek status pada standby

SQL> SELECT NAME, OPEN_MODE, DATABASE_ROLE FROM V\$DATABASE;

-- cek services mrp

```
SQL> SELECT PROCESS, STATUS, THREAD#, SEQUENCE# FROM V$MANAGED_STANDBY;
```

-- cek gaph pada replikasi

```
SQL> alter session set nls_date_format='DD-MM-YYYY HH24:MI:SS';
```

```
SELECT A.THREAD#, B.LAST_SEQ, A.APPLIED_SEQ, A.LAST_APP_TIMESTAMP, B.LAST_SEQ-  
A.APPLIED_SEQ ARC_DIFF
```

```
FROM (SELECT THREAD#, MAX(SEQUENCE#) APPLIED_SEQ, MAX(NEXT_TIME)  
LAST_APP_TIMESTAMP
```

```
FROM GV$ARCHIVED_LOG WHERE APPLIED = 'YES' OR APPLIED='IN-MEMORY' GROUP BY  
THREAD#) A,
```

```
(SELECT THREAD#, MAX(SEQUENCE#) LAST_SEQ FROM GV$ARCHIVED_LOG GROUP BY THREAD#)  
B WHERE A.THREAD#=B.THREAD#;
```

-- Cek proses apply di STANDBY

```
SELECT PROCESS, THREAD#, SEQUENCE#, STATUS, BLOCK#, BLOCKS
```

```
FROM V$MANAGED_STANDBY
```

```
WHERE PROCESS LIKE 'MRP%';
```

Alternatif Monitoring (Manual Gap Check)

Kalau kamu mau pantau secara manual, kamu bisa lakukan:

-- Di primary:

```
SQL> SELECT MAX(SEQUENCE#) FROM V$ARCHIVED_LOG;
```

-- Di standby:

```
SQL> SELECT MAX(SEQUENCE#) FROM V$ARCHIVED_LOG WHERE APPLIED = 'YES';
```

-- set archive_dest 2 untuk mengirim archive dari primary

```
ALTER SYSTEM SET LOG_ARCHIVE_DEST_2 ='SERVICE=orcl_primary ASYNC  
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=orcl_primary'  
SCOPE=BOTH;
```

```
ALTER SYSTEM SET LOG_ARCHIVE_DEST_2 ='SERVICE=orcl_standby ASYNC  
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=orcl_standby'  
SCOPE=BOTH;
```

-- DBWMS --

```
ALTER SYSTEM SET log_archive_dest_2='SERVICE=dbwms_primary LGWR ASYNC  
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=dbwms_primary'  
SCOPE=BOTH;
```

```
ALTER SYSTEM SET log_archive_dest_2='SERVICE=dbwms_standby LGWR ASYNC  
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=dbwms_primary'  
SCOPE=BOTH;
```

NOTES MENGENAI KONFIGURASI BESERTA PENJELASANNYA

----- Konfigurasi di Primary

1. LOG_ARCHIVE_CONFIG='DG_CONFIG=(orcl_primary,orcl_standby)'

Fungsi: Menentukan database mana saja yang termasuk dalam konfigurasi Data Guard.

Keterangan: Parameter ini memungkinkan komunikasi antar anggota Data Guard, yaitu antara orcl_primary dan orcl_standby.

2. LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/archive
VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=orcl_primary'

Fungsi: Menentukan lokasi lokal untuk menyimpan archive log di primary database.

VALID_FOR: Berlaku untuk semua jenis log (redo dan archive) dan untuk semua peran (primary atau standby).

DB_UNIQUE_NAME: Harus cocok dengan nilai DB_UNIQUE_NAME dari database ini.

3. LOG_ARCHIVE_FORMAT='%t_%s_%r.arc'

Fungsi: Format nama file untuk archive log.

Penjelasan:

%t = thread number

%s = log sequence number

%r = resetlogs ID

4. LOG_ARCHIVE_MAX_PROCESSES=4

Fungsi: Menentukan jumlah maksimum proses arsitektur log.

Keterangan: Disarankan > 1 untuk performa lebih baik saat banyak archive log yang perlu dikirim.

5. DB_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'

Fungsi: Otomatis mengonversi path datafile dari primary ke standby.

Contoh: File di /u02/oradata/orcl_primary akan dikonversi ke /u02/oradata/orcl_standby saat standby dibangun dari primary.

6. LOG_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'

Fungsi: Sama seperti di atas, tetapi untuk logfile (redo log).

Tujuan: Supaya standby tahu lokasi baru untuk menyimpan redo log.

7. STANDBY_FILE_MANAGEMENT=AUTO

Fungsi: Mengelola file secara otomatis di standby saat ada perubahan di primary.

Contoh: Jika ada datafile baru di primary, standby otomatis membuatnya.

8. FAL_SERVER=orcl_standby

Fungsi: Menyatakan ke mana primary akan meminta archive log yang hilang (jarang dipakai di primary, tapi tetap di-set untuk completeness).

9. FAL_CLIENT=orcl_primary

Fungsi: Nama service yang digunakan oleh primary saat menerima log gap request dari standby.

10. REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE

Fungsi: Mengaktifkan penggunaan password file untuk koneksi antar database (misal untuk redo transport).

EXCLUSIVE: Setiap DB memiliki password file sendiri, dan bisa digunakan untuk sysdba remote login.

----- Konfigurasi di Standby

1. DB_UNIQUE_NAME='orcl_standby'

Fungsi: Memberi nama unik untuk standby database, wajib berbeda dari primary.

2. LOG_ARCHIVE_CONFIG='DG_CONFIG=(orcl_primary,orcl_standby)'

Fungsi: Sama seperti di primary, agar standby mengenal anggota DG lainnya.

3. LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/archive
VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=orcl_standby'

Fungsi: Menentukan lokasi lokal untuk archive log di standby.

4. LOG_ARCHIVE_DEST_2='SERVICE=EXAPRIMARY ASYNC
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=orcl_primary'

Fungsi:

Menentukan tujuan pengiriman redo log.

SERVICE=EXAPRIMARY: Net service name untuk primary.

ASYNC: Gunakan mode asynchronous (umumnya lebih cepat, tapi dengan risiko kecil keterlambatan).

VALID_FOR: Hanya mengirim online redo log jika berperan sebagai primary (ini relevan saat role switch).

5. LOG_ARCHIVE_FORMAT='%t_%s_%r.arc'

Sama seperti di primary, untuk format nama archive log.

6. LOG_ARCHIVE_MAX_PROCESSES=4

Sama seperti di primary, mengatur jumlah proses untuk archive.

7. DB_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'

Fungsi: Mengonversi path datafile dari primary ke standby secara otomatis.

8. LOG_FILE_NAME_CONVERT='/u02/oradata/orcl_primary','/u02/oradata/orcl_standby'

Sama dengan di atas, tapi untuk redo log file.

9. STANDBY_FILE_MANAGEMENT=AUTO

Agar datafile di standby dikelola otomatis saat terjadi perubahan di primary.

10. FAL_SERVER=EXAPRIMARY

Fungsi: Jika standby mendeteksi log gap, dia akan meminta archive log ke primary menggunakan EXAPRIMARY.

11. FAL_CLIENT=EXASTANDBY

Fungsi: Nama service yang digunakan standby saat mengidentifikasi dirinya sendiri saat meminta log ke primary.

12. REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE

Sama seperti di primary, penting untuk authentication antar database via redo transport.